Forklift: a humanoid or a driving-robot kit drives a 2.5 t counterbalance truck through its own controls

Gym Anything for Robotics

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A Unitree H1-2 with Inspire hands sits in the operator's seat of a 2.5 t LPG counterbalance forklift and does the industrial-truck operator's job with its own hands and feet: start the truck, release the parking brake, steer with the spinner knob, feather the accelerator, inching and brake pedals, select a direction, work the lift, tilt and side-shift levers, sound the horn, set the brake and switch off. The second operator offered is a vehicle-testing driving-robot kit bolted into the cab: a steering robot, pedal robots and lever, selector, hand-brake, key and horn actuators, each held to its product's published force and speed. Nothing drives the truck except contact between the operator and the truck's controls. The truck is NVIDIA's Isaac Sim Forklift C mesh re-rigged as a Hyster H2.5A (masses, axle loads, drivetrain, lift and lowering speeds fitted to the manufacturer's sheet) with super-elastic tyres, a torque-converter powershift drivetrain, a hydraulic drum service brake, hydrostatic steering and staged mast hydraulics; the loads, racks, conveyor, dock levelers, sprung trailers and the yard ramp are explicit bodies. Newton 1.5 with MuJoCo on the CPU at 0.5 ms; Isaac Sim 5.1 RTX path tracing in a render-only worker for the cameras.

This package builds the environment and demonstrates that it supports its interactions. Its demonstrations are labelled ASSISTED (phantom forces at the controls, privileged state), ASSISTED ROBOT FIXTURE (the H1-2's own hand or foot on a control, its pelvis supported and the limb guided by bounded external forces), SCRIPTED ROBOT (the H1-2's own motor channels, IK with privileged state) or SCRIPTED KIT (the kit's own actuator channels, driven by a script with privileged state). None of them is a policy. The tasks are meant to be hard.

Status. All fourteen tasks are completed through the unmodified evaluator by an assisted autopilot at the truck's controls. Seed 0 of every task and a second seed of each task whose target varies make 27 runs. 25 score 1.00. The two load-bales runs were scored 0.909 by an earlier version of the evaluator, which read a bale resting against the nose wall as outside the trailer, and replays on the current code succeed (section 11). Eighteen more put-away seeds all succeed. Five of seven predeclared negative controls fail as predicted, and two predictions are refuted and kept. In assisted fixtures (its pelvis supported, the working hand or foot guided by a bounded external force, each control moved only by the contact of its own links) the seated H1-2 works the lift and tilt levers, the accelerator and the service brake, and in an earlier build it also worked the direction stalk and the horn and released the parking brake. It does not apply the parking brake again, turn the key, press the inching pedal, pull the side-shift lever its last percent or turn the wheel through repeated grips (section 10). Without assistance the scripted robot passes none of the controls by its own rule. The driving-robot kit works every control by contact. At the products' published forces it takes the brake pedal to 78 % and the inching pedal to 87 % of their travel, and its brake press reaches 1.5 times the line pressure of the braking standard's pedal force. Driven by a script with privileged state it completes start-and-park on two seeds and aisle-cruise on one, scored by the unmodified evaluator (section 4.4). Every start state stays at rest for two minutes. Eight seeds of violent legal commands run 60 s each without a fault at the ledger's finger inertia (in an earlier build, one of two seeds faulted at the low end of that value's open range). Open, and said plainly in section 15: the parking brake's re-apply and the robot's own operation of the truck, a trailer walk under repeated entries (the model does not produce it), the sheet's gradeability (missed both ways), the laden 0-15 m time (13 % too brisk), a laden stop at the standard's pedal force that lifts the steer wheels for 0.57 s, and lever slams with the rated load lifting the steer axle (physics, failed by the evaluator).

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1. The real job and what this environment covers

The action group "steer forklift" collects 18 task statements from O*NET whose leaf action is steering a material-handling vehicle; the grounding release allocates them $14.25 B of annual paid-work value. 97.6 % of that value is two jobs done with one kind of truck: moving loads between loading, processing and storage areas inside a facility (76.3 %, led by Industrial Truck and Tractor Operators, 53-7051) and loading trucks and trailers at docks (21.3 %, led by Recycling and Reclamation Workers loading bales, 53-7062.04). The remaining rows (a track-laying vehicle, a highway sweeper, an underground shuttle car, an amusement-park vehicle) are 2.4 % and are not modelled.

The job as written in the operating rules (OSHA 29 CFR 1910.178 and its eTool, read): inspect, sit, belt, start; travel with the forks 10–15 cm above the floor and the mast tilted back; stop 8–12 in short of a pallet, square up, level the forks, inch in, lift about 4 in, tilt back, back out; slow down and sound the horn at cross aisles; on grades keep the load upgrade; place the load on a floor, rack beam, conveyor or trailer floor, lower, back out; park with the forks lowered, controls neutral, the power off and the brake set. At a dock the trailer must be chocked or held by a restraint. The truck class that does this work is the 2.5 t LPG counterbalance truck with a pivoting steer axle (the "stability triangle"), whose dominant fatal failure is a lateral overturn in a turn (NIOSH Alert 2001-109: overturns 22 % of 1,021 deaths; INRS TI-NT-46).

What the environment covers: the full operator's workspace of that truck (pedals, levers, stalk, knob, key, horn, parking brake, seat switch, lap belt), three facility presets (a racked warehouse with a conveyor, a four-door shipping dock with levelers and trailers, a paved yard with a mobile ramp into a trailer) and 14 tasks across them (section 7). What it does not cover: pedestrians and other traffic, the pre-shift inspection, refuelling, attachments other than the side-shifter, electric or narrow-aisle trucks.

2. Scene at real scale

2.1 Physical geometry

- **Truck.** The Forklift C asset's meshes, measured: wheelbase 1.649 m (H2.5A sheet 1.650 m), front tyres 7.00-12 (Ø 0.646 m), rear 6.00-9 (Ø 0.513 m), overhead guard slat tops 2.111 m in the asset; fork blades 102 mm wide; the fork shanks' load face 0.545 m ahead of the front axle (`geometry.py`, `params.py`). The truck is 21 rigid bodies on joints (chassis, oscillating steer axle, two knuckles, four wheels, mast outer/intermediate/inner stages, carriage, side-shifter with forks, steering wheel, spinner knob, horn button, three pedals, three hydraulic levers, direction stalk, parking-brake lever, key and the suspended seat). Collision geometry is convex hulls of the asset's parts without its decal, glass and lamp meshes (the outer upright's sticker mesh would otherwise make a phantom lug ahead of the mast), except each fork blade's flat run: eight 125 mm box primitives with the blade's measured top (0.1402 m), the tapered tip a hull; the cab interior where the robot's legs and hands work is a CoACD decomposition of the asset's cab mesh with the footwell carved out and its floor, front wall and toe board authored as plates at the asset's surfaces (`geometry.py`). Which truck this is, value by value, is section 3.1.
- **The overhead guard at the sheet's height, and the operator's head.** The guard is raised 0.049 m so that its slat tops stand at the H2.5A sheet's h6 of 2,160 mm (`params.GUARD_RAISE`). The seated H1-2 starts leaning 5° forward (`operator.START_LEAN`, inside ISO 11226's 0-20° trunk band), which puts its head under the guard's open bay: at rest its head top is 2.074 m, 28.5 mm under the rear slat's underside. Without the lean a hard stop drove the head into the rear slat at 3.1 kN. Every task fails if any part of the robot meets the guard with more than 130 N, ISO/TS 15066's limit for the skull and forehead (`params.HEAD_CONTACT_LIMIT`, section 7). The cushion at its bottom stop is 1.036 m (40.8 in) below that slat, over the 39 in seat-to-guard minimum of OAR 437-002-0227. The seat sits fully forward on its 150 mm rails (75 mm ahead of centre, as the tall H1-2 needs), and the seat foam is set 30 mm under the operator (`SEAT_RAIL`, `SEAT_FOAM_SET`). Hyster's low-profile seat option was not used: no lowering figure is published and this cab has 34 mm under the cushion.

- **Warehouse** (`facility.py`): 50 × 36 m, three selective rack rows (a single row on the west wall, a back-to-back double row and a single row; 42 in frames, 96 in beams, 3 in uprights, 16 ft frames, beam levels 1.60 and 3.20 m, seven bays per row, the double row's frames on 12 in spacers leaving a 6 in flue between its loads) on two working aisles 4.50 m wide between the stored loads' faces. The beam faces stand 4.65 m apart, because the loads overhang them by 3 in. The VDI 2198 right-angle stacking aisle Ast = Wa + x + l6 + a for this truck with a 48 in pallet carried lengthways is 2.365 + 0.545 + 1.219 + 0.200 = 4.329 m (Wa the outside turning radius measured in E4, x the fork face, l6 the pallet, a the 200 mm that Hyster's guide allows), so the counterweight keeps 271 mm (`params.AISLE_RACK`). Stored unit loads stand in 80 % of the rack locations, 500-1,000 kg each. Their heights are drawn from 0.9-1.2 m, so each fits its opening with a 6 in lift-off clearance under the beam above (`params.RACK_LIFT_OFF`: the top of Interlake Mecalux's 4-6 in range and Cisco-Eagle's minimum). Around the put-away task's target slot the stored unit loads are bodies resting on the beams by contact (every level of the slot's face in its bay and the bays either side, and the same bays at the slot's level across a double row's flue); elsewhere they are static structure. A main aisle with two marked cross intersections, painted floor bays, receiving lanes, staging lanes, a parking lane with stop lines, a powered pallet roller conveyor (41.0-47.2 m) feeding a press, columns with guards. Static structures are watched for strikes.
- **Dock**: four 9 × 10 ft doors on 12 ft centres at a 48 in dock face, hydraulic levelers (6 ft 6 in × 8 ft deck, 16 in lip) in pits, 4.5 in bumpers, 53 ft dry-van trailers (Wabash DuraPlate HD inside dimensions) spotted against the bumpers, staging lanes and a bale row.
- **Yard**: asphalt, a dock apron and building wall, a mobile yard ramp (Bluff MobilePRO class: 30 ft incline, 6 ft level-off, 84 in wide, 8 in curbs, two 4 × 10 in bumpers against the trailer's rear frame) whose level-off is set one lip-plate thickness above the trailer floor, and a multi-piece hinged lip: four 16 in plates, each pinned to the level-off and free between 3 in below in-line and upright (manual: "adjustment up to 3 in up and 3 in down"), lowered onto the trailer floor and resting on it by contact as the sprung trailer sinks and rises. The spotted trailer stands on its chocked tandem, its landing gear and two stabilizing jacks under its nose (OSHA 1910.178(m)(10); the ramp's manual forbids use without them), cranked up to touch with no preload and pushing only when pressed. Parked trailers, a marked 4 m lane and a painted drop zone.

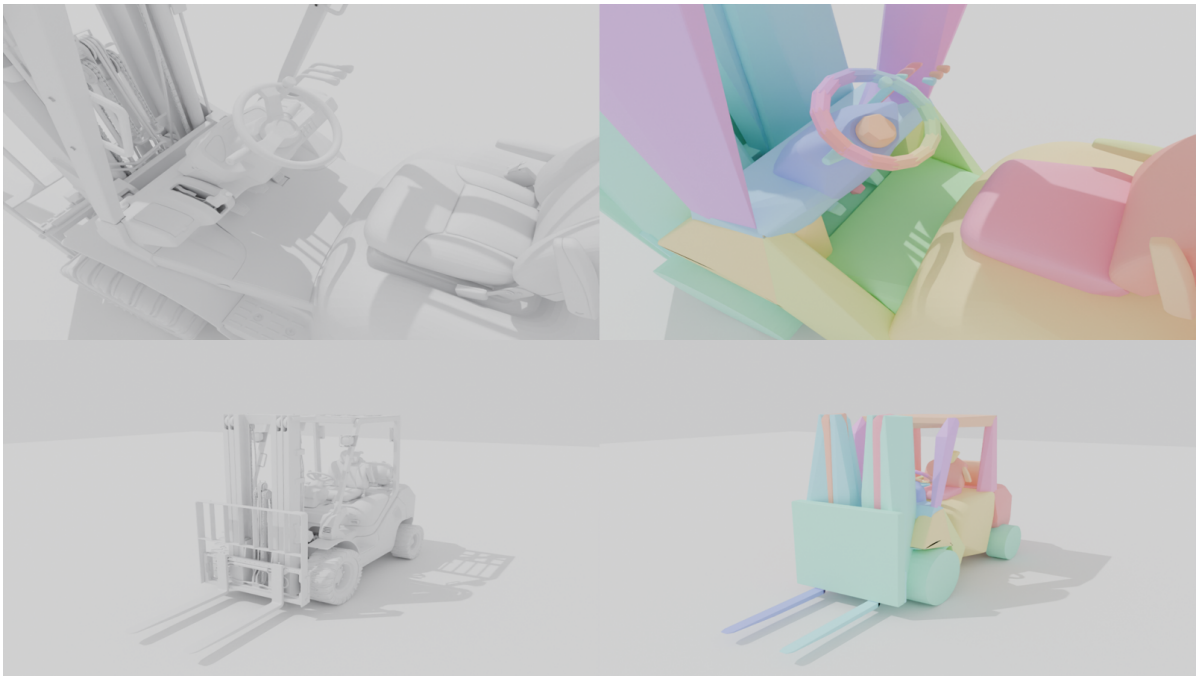


Figure 1. The truck's visual meshes (left) and its collision shapes (right): the cab interior and the front quarter.

2.2 Appearance and cameras

The render-only worker draws the truck from the asset's own meshes and materials, the robot from its URDF meshes, the layout from its simulated geometry with PBR materials, and dresses the buildings with NVIDIA's Simple_Warehouse props (roof beams, high-bay lamps, cartons and barrels along walls, fuse boxes, aisle signs, cones, a wet-floor sign beside a wet patch). Lighting is physical: the warehouse and dock are lit by 1.2 × 0.4 m high-bay LED panels on a 6 m grid at 8.4 m, 1.0×10^4 nits each (15,000 lm, a 150 W high bay, over the diffuser, $L = \Phi/\pi A$), which puts about 390 lx on the floor (IES: 108 lx for active bulky storage, 323 lx for small items and shipping and receiving, 323–538 lx at docks; search), with a weak uniform fill; the camera exposes ISO 400, f/2.8, 1/120 s (EV100 7.9). The yard is lit by an HDR sky (partly cloudy; the dome scaled so the ground receives about 40 klx) and exposed ISO 100, f/8, 1/250 s (EV100 14).

Trailer interiors have no light of their own: each dock door has an LED dock light on a swing arm (2,500 lm, aimed into the trailer) and the truck carries two LED work lights on the front of its overhead guard (1,500 lm each, aimed forward and down; render only, luminance from flux as for the lamps; flux values are estimates of the class). Every camera meters its own exposure from its frames, as the H1-2's head camera (a RealSense D435i, auto-exposing by default) and a human eye do, so the head camera's image stays readable inside a trailer.

Cameras: the policy camera is the selected operator's. For the H1-2 it is its head camera, a RealSense D435i colour stream (848 × 480, 69.4° horizontal, 30 Hz frame clock). For the driving-robot kit it is a D435i-class cab camera with the same image, on a mast hanging from the overhead guard at an operator's eye point and looking forward 15° down. Diagnostic cameras (owner key only): an overview, a chase camera above the counterweight looking forward over the overhead guard, a cab camera, a fork camera on the carriage, a side camera, a pedal camera low on the right of the seat looking into the footwell, a hands camera, and two cab cameras from the left and the right. The evidence films add their own cameras: chase cameras that follow the truck and, on every frame, take the first of their placements (front quarter left or right, beside the truck, high behind) whose line of sight from the truck is clear of racks, walls and stored loads, hold it for at least a second of film, never aim at a point inside a load, and pull in where something still stands in the way; cameras fixed inside a trailer's nose and behind its doors; and, for the robot fixtures, close-up cameras fixed to the truck and framed on the working hand or foot.

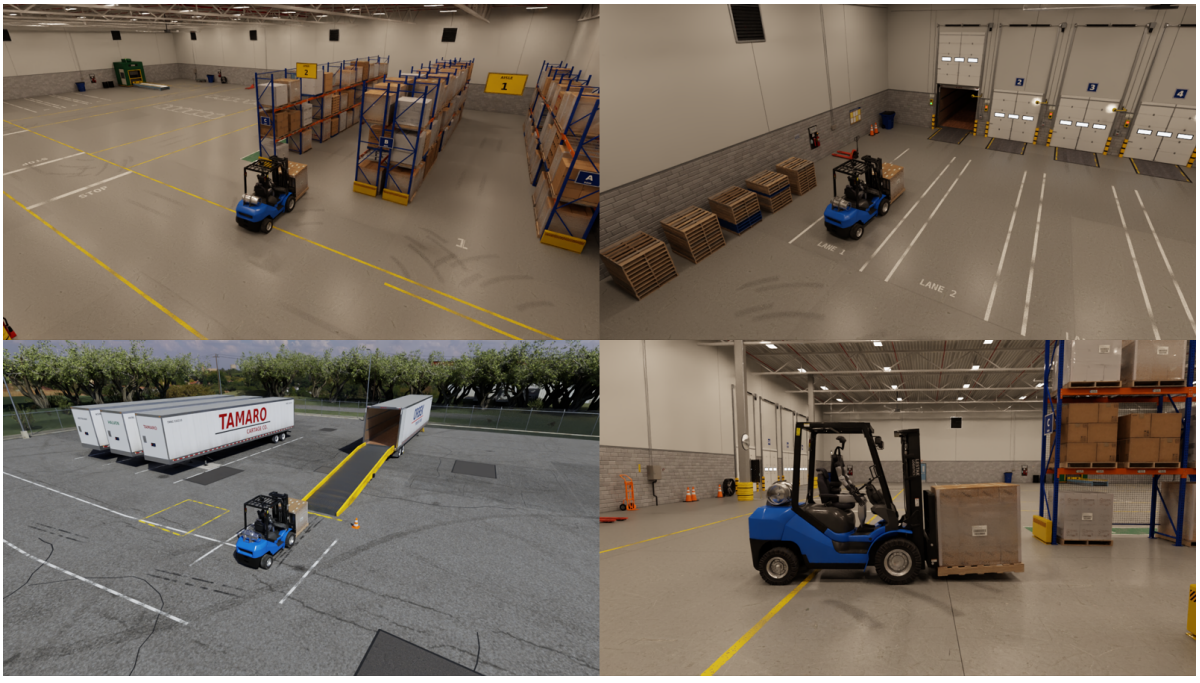


Figure 2. Start states, rendered from the physics state: the warehouse (put-away-rack), the dock (load-trailer), the yard (ramp-up-loaded), and the truck from the side with its load and the seated H1-2.

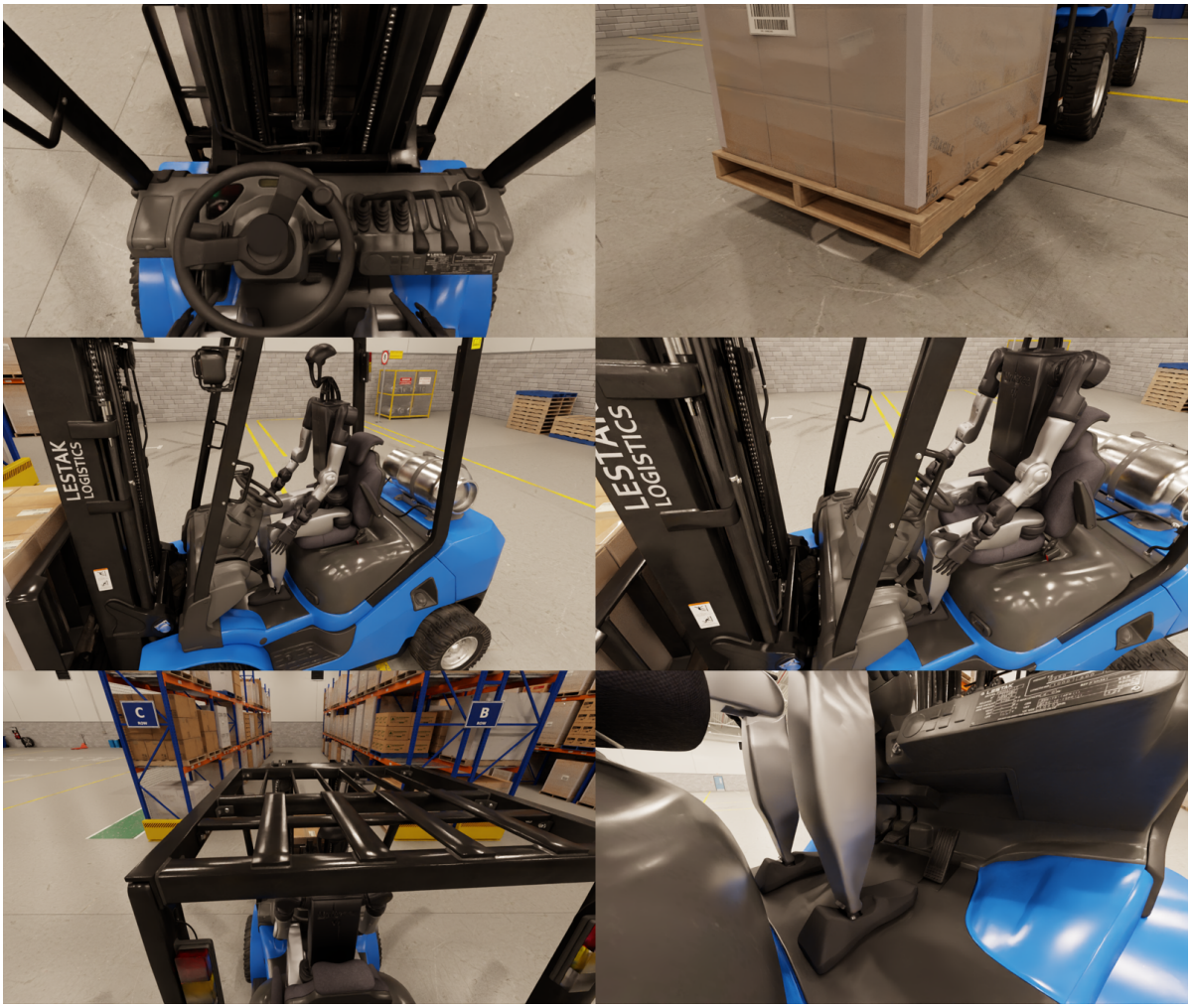


Figure 3. The cameras at the warehouse start state: the policy's head camera; the fork camera; the cab and hands cameras; the chase camera above the counterweight; the pedals camera in the footwell.

3. The truck: mechanisms and equations

3.1 Which truck this is

The simulated truck is a hybrid, and every value below says which vehicle it comes from. Its body is NVIDIA's Forklift C asset: a 2.5 t-class LPG counterbalance truck with a 3-stage full-free-lift mast and a side-shifter. Its masses, drivetrain and hydraulic speeds are the Hyster H2.5A LPG's, from a sheet whose standard truck has a 2-stage limited-free-lift mast and no side-shifter. Its steering components are taken from electric trucks' documents.

What	Value here	Source vehicle
Wheelbase, tyre sizes, mast, side-shifter, cab, guard	measured from the meshes	Forklift C asset
Overall width / rear tread	1.322 m / 1.136 m (H2.5A: 1.160 / 0.958 m)	asset
Fork face ahead of the front axle	0.545 m (H2.5A: 0.470 m)	asset
Outside turning radius	2.365 m, measured E4 (H2.5A: 2.263 m, PrimeMatch 2.357 m): about 4.5 % wider than the sheet's truck	asset geometry + steering below
Service weight, axle loads, centre of gravity	3,990 kg; 1,728 / 2,262 kg unladen	Hyster H2.5A LPG, distributed over the asset's parts

What	Value here	Source vehicle
Data plate	the H2.5A's rated moment (2,500 kg × 0.970 m) over this fork face: 2,320 kg at 500 mm	derived; Hyster rates its H2.5A with a 3-stage FFL mast of this lowered height and side-shift at 2,310 kg (p. 9)
Mast stage stroke	1.69 m per stage	Hyster's 3-stage FFL table for the asset's 2.17 m lowered height (p. 9)
Engine, converter, reduction, travel and lift speeds	Yanmar 4TN88G curve; ratios fitted	Hyster H2.5A LPG sheet
Steering: turns lock to lock, metering unit, relief, wheel angles, slip at the stops	4.8 turns; 80 cm^3/rev; 12 MPa; 80° inner; 3.5 rpm	Crown FC 5200, Hangcha CPD20-35 (electric trucks), Danfoss OSP, Eaton US 6,769,249 B2
Service brake: pedal ratio, master and wheel cylinders, drums, shoe clearance; pedal heights and plays	6.3; 19.05 / 28.58 mm; 310 mm duo-servo; 0.425 mm	TCM FG20T3/FG25T3 service manual (2.0-2.5 t LPG)
Tyres	measured 18x7-8 SE laws scaled to 7.00-12 and 6.00-9	Oh et al. 2013, SUPREM, Pross & Bruns 2019; Trelleborg sizes

So the truck is "a 2.5 t-class LPG counterbalance truck built on the Forklift C, with the H2.5A's masses and drivetrain", not an H2.5A. The asset's longer fork-face lever is why its data plate is 2,320 kg, not 2,500 kg, at 500 mm; the mast's and side-shifter's masses are the H2.5A's estimates spread over the asset's stages (`params.M_*`), not a re-derivation for a triplex.

3.2 Masses

Service weight 3,990 kg and axle loads 1,728 / 2,262 kg unladen, 5,698 / 792 kg with 2,500 kg at 500 mm (Hyster H2.5A, read). The sheet is self-consistent (the unladen axle loads put the centre of gravity 0.935 m behind the front axle; moving 2,500 kg to 0.970 m ahead predicts 792.3 / 5,697.7 kg laden). Body masses are distributed over the asset's parts to reproduce the unladen axle loads and centre-of-gravity height (`params.M_*`, `CG_*`); the measured static axle loads of every start state are in section 12.

3.3 Tyres (`tyres.py`)

Super-elastic tyres are sampled as 3 ribs across the width × an arc of points on the lower circumference and tested against drivable surfaces (oriented rectangles fixed to the world or to a moving body: floor zones, ramps, leveler deck and lip, the yard ramp's lip plates, trailer floors). Per rib, with the deepest penetration p and a Maxwell element's internal state s:

F_rib = (K/3) p^1.5 + k_s (p - s), ds/dt = (k_s / C)(p - s), k_s, C ∝ (F / 10 kN)^0.28

K is fitted to the load-deflection curve Oh et al. (2013) measured on an 18x7-8 super-elastic forklift tyre (7.1×10^6 N/m^1.5) and scaled to the 7.00-12 front (8.2×10^6) and 6.00-9 rear (6.0×10^6) tyres by section width × $\sqrt{}$radius. The damper is a Maxwell element (dashpot C in series with spring k_s: 14.5 kN·s/m and 272 kN/m at a 10 kN wheel load) fitted to the same paper's measured damping (13.2 kN·s/m at 1 Hz, 3.8 at 5 Hz), growing with wheel load as the paper's 1 Hz values do. Tangential forces are a brush (relaxation) model whose steady state is the SUPREM law for forklift tyres (read):

du/dt = -v_slip - |v_x| u / σ, σ = max(σ_min, |v_x| T), T = k_d (v [km/h])^-k_v F_y = F_z μ_B exp(-F_z/k_F1) tanh(atan(u_y/σ)[deg] / (k_a + k_F2 F_z)) F_x = F_z μ_B ρ exp(-F_z/k_F1) tanh((u_x/σ_x) / κ*), σ_x = max(σ_min, a), a = $\sqrt{}$(2 r p)

μ_B is SUPREM's base coefficient of the surface. For dry sealed concrete it is the measured drum level of the tyre whose SUPREM coefficients the law uses: its quasi-static lateral coefficient is 0.711 at 10 kN, so μ_B = 0.711 exp(10/50.9) = 0.865. The drum is coated to reproduce tarmac friction, and Wong gives tyres the same peak on dry asphalt and dry concrete; no measurement of a super-elastic tyre on a warehouse floor was found, so dry concrete is taken as the drum (estimate). The longitudinal peak is $\rho = 1.227$ times the lateral one: on the same drum a super-elastic tyre at 10 kN peaks at 0.94 braking and 0.766 sideways (Pross & Bruns 2019), against this law's 0.93 and 0.757 for the front tyre. The braking-slip curve of that paper sets the slip scale $\kappa^* = 0.22$. The longitudinal deflection relaxes over the contact patch's half-length (the brush model's relaxation length with a rigid carcass), not over the lateral σ: with the lateral σ the braked front wheels juddered at 5.8 Hz in a 300 N stop. The resultant

is capped on Kamm's ellipse. Once the patch slides, the cap drops to 0.904 of the peak (the locked-wheel 0.85 over the peak 0.94 of the same curve) until the bristles ask for less than that again (Coulomb stick-slip: break away at the peak, slide at the locked value). The other surfaces keep estimated ratios to dry concrete: wet 0.594, dusty 0.646, steel plate 0.542, grating 0.697, oak 0.594. The lateral force builds with SUPREM's first-order delay and at standstill u is a bristle spring that holds a parked truck with a fixed elastic set instead of creeping. A bristle damper acts at low speed; a torsional bristle carries turn slip at standstill (capped at μ F_z r_t); rolling resistance acts on the wheel joint; the outer tyre's patch shift gives a moment F_y/k_M. The normal and lateral forces act at the contact point; the longitudinal force acts at the effective rolling radius r_e = r − p/3 (Wong), where its slip is measured, so the wheel torque is F_x r_e and the tyre's power balances; rolling resistance is the wheel torque C_rr F_z r_e. Each acts equal and opposite on the surface's body (the trailer is pushed by the truck's braking; the ramp's lip plates are loaded by the wheels). The truck starts at rest on its tyres (`world._spawn_on_tyres`). Each rib bears on the deepest surface beneath it, at the deepest of its arc points and, per surface, the rib circle's exact deepest point, so on a grade the normal force passes through the axle; a penetration deeper than 0.10 m is a surface far above the tyre, not contact. Wheel shapes never touch drivable surfaces through MuJoCo; they still strike curbs, pallets and racks. Contact between other bodies and floors or structures uses its own friction (`MU_STRUCTURE` 0.50, the wood-on-concrete value, so that MuJoCo's larger-of-the-pair rule leaves each load its own coefficient).

3.4 Drivetrain, service brake and inching (`drivetrain.py`)

engine (governed torque curve, 174 N·m at 1,690 rpm, 42 kW; idle 750 rpm; starter) → torque converter (K-factor capacity, stall ratio 2.94 falling to coupling at speed ratio 0.85, stall 1,900 rpm) → turbine → forward/reverse clutch packs (capacity ramped by engagement and cut by the inching pedal) → reduction 16.43 × efficiency → open differential → front wheels. Stall ratio and reduction are **fitted** together to the sheet's drawbar pull (17,145 N at 1.6 km/h laden) and top speed (18.7 km/h unladen) at the tyres' effective rolling radius, so those two numbers check the fit; the laden top speed, both 0-15 m times and the sheet's gradeability are held out (section 9.1). The hydraulic pump loads the engine with its shaft torque D p / (2π η_hm) (η_hm 0.88; the open-centre circulation pressure 0.5 MPa when no function is selected; the lift circuit's pressure while raising, the tilt or shift column's when driven, the steering pressure when it is higher), so lifting a load at idle pulls the engine down (E5). Engine and turbine speeds are scalar states; the clutch is a smooth Coulomb coupling; the converter couples engine and turbine by linearised backward Euler. The truck creeps at idle in gear, as a torque-converter truck does.

The service brake is hydraulic and worked by force. Its hardware is a 2.0-2.5 t LPG truck's (TCM FG20T3/FG25T3 service manual: pedal ratio 6.3, master cylinder 19.05 mm, wheel cylinders 28.58 mm, duo-servo drums 310 mm, 0.4-0.45 mm of shoe clearance). The pedal pushes the master cylinder through its ratio. The oil first moves the shoes onto the drums (1.09 cm^3) against their return springs (0.4 MPa), and beyond that the circuit stiffens with pressure: V − takeup = a ln(1 + (p − p_shoe)/p_c). Two anchors set a and p_c: the braking standard's 450 N leaves the pedal at 70 % of its travel (an authored reserve: no pressure-volume curve of a forklift brake was found), and the pedal reaches its floor stop only at 1,800 N, the force ISO 6292's strength clause has a fully pressed pedal withstand. The pedal feels the line pressure back through the same ratio, so the pressure follows the force on the pad and the travel follows from the oil the brakes take. The fluid's flow through 2 m of 3/16 in brake tube damps the pedal. Each front drum's torque is the brake factor × drum radius × the wheel cylinder's force above the shoe springs. The brake factor, 2.65, is **derived**, not fitted: it is the smallest value that meets the service-brake standard with this hardware, a drawbar drag of 25 % of the truck's weight plus rated capacity (OSHA's 1990-06-21 interpretation of B56.1) at a pedal force of 450 N (ISO 6292:2020, read through GB/T 18849-2023). A duo-servo drum's real factor and this truck's margin over the standard are not published (open). Service and parking brakes act as dry-friction stick/slip elements on the front-wheel joints (a stiff spring anchored where the wheel stuck, capped at the brake torque), so a braked wheel holds at rest without creep, and every joule of slip is booked. The parking brake holds 3,000 N·m per wheel (open: the laden truck on 15 % needs 1,460 N·m).

The inching pedal follows TCM's design, transferred to the asset's pedals as fractions of the travel. Over the first 17 % of its travel it cuts the clutch packs' capacity linearly to zero. There its lug meets the brake pedal (Komatsu's US4441596A couples the pedals the same way), and past that point it brakes only by pushing the brake pedal, so the foot then carries the brake's force too. Its rest is turned back 8.7° about its pivot so that its pad stands at the brake pad's height, as TCM draws the two pedals.

3.5 Steering (`steering.py`)

Steering wheel → orbitrol spool and meter (80 cm^3/rev) → cylinder → left knuckle; the right knuckle follows by an Ackermann equality, held as stiff as the steel linkage between them (a 2 ms equality time constant). The unit is a non-reaction type (Danfoss OSPB/OSPC): the cylinder's pressure never turns the wheel. The wheel turns the spool against its neutral setting springs (0.8 N·m preload rising to 3 N·m, Danfoss) up to a deflection limit of ±9°. With pump flow the open spool meters flow in proportion to its opening, up to the priority flow (14 L/min at rated speed), D per meter revolution into the cylinder, minus leakage proportional to the cylinder pressure (so knob and wheel angles drift apart with every hard turn). The leakage is derived from the slip a vehicle maker allows against the stops, 2-5 rpm of the wheel (Eaton US 6,769,249 B2): the middle, 3.5 rpm at the 12 MPa relief. With the engine off the meter turns only when the wheel drives it through the stop, as a hand pump, and the wheel feels the cylinder pressure as torque $\Delta p\ D\ /\ 2\pi$; E4 checks that the engine-off truck steers within the 120 N·m an average operator can exert (Danfoss). The cylinder is a stiff oil column capped by the 12 MPa steering relief; at the stops the oil does not move. The kingpins are the axes of the asset's knuckle bosses, measured in E21, and the knuckles stop at −80° and 56.2° (Hangcha: inner 80°, outer 55.7°). One wheel turn moves the knuckles about 28°; a left turn of the wheel steers the rear wheels right and the truck left. The lock-to-lock turns (4.8, Crown FC 5200) set the cylinder's volume per degree of knuckle, so reproducing them checks that fit.

3.6 Mast hydraulics (`hydraulics.py`)

Gear pump (32.7 cm^3/rev, **fitted** to the 0.60 m/s laden lift at 2,500 rpm) → priority valve (steering first) → three-spool open-centre valve worked by the cowl levers → lift circuit (two main lift cylinders, a free-lift cylinder on the carriage at 2:1), two double-acting tilt cylinders, the side-shift cylinder; 18.5 MPa main relief at the pump; lowering regulator. Each cylinder is a compressible oil column, force k (x_oil − x) with $k = \beta\ A^2\ /\ V$. Each lift cylinder reaches the valve through its own line, 3/4 in bore and 3 m (main) or 5 m (free lift) long, whose pressure drop is Darcy friction plus the fittings' losses (estimates). The lift columns carry no pressure cap of their own (the relief bounds the pump's pressure while raising; a closed cylinder's pressure is whatever its load makes it), and the tilt and shift columns carry a 0.15 damping ratio. Mast friction (rollers, chain sheaves, cylinder seals) is Coulomb, 6 % of the force each lift column transmits (estimate; open 3-10 %). In the lift circuit the node pressure solves the flow balance Σ q_line,i(p_i − p_n) = Q(p_n) by bisection, so the cylinder needing the lower pressure moves first on raising (free lift, then the main stage) and the higher-pressure one first on lowering; a load whose pressure exceeds the relief never lifts. The last 50 mm of each lift cylinder's stroke is throttled (a cubic taper of the line's opening to 1 %), as cylinder end cushions are, so the stages do not slam into their stops. The stage stroke is 1.69 m (Hyster's 3-stage table, section 3.1). Lift-cylinder leakage (**fitted**) lets a raised rated load creep down about 35 mm in 10 min (ISO 3691-1 allows ≤ 100 mm). Lowering works with the engine off; tilt and shift lock.

The lowering regulator is a load-compensated pressure compensator: it holds the drop across the lift spool's own lowering orifice at 1.70 MPa, so the lowering speed follows the lever whatever the load, as the flow-rate valve of Toyota Industries' EP2962981A1 does. Its setting and its opening are **fitted** to the sheet's 0.58 and 0.50 m/s laden and unladen lowering speeds; its 0.05 s lag is open. Whether the H2.5A lowers this way or through a restrictor regulator that caps a spool-metered flow is not published (open, section 15). An open seat switch blocks all spools.

3.7 Controls and the operator-presence system (`controls.py`)

Every pedal, lever, the stalk, the parking-brake lever, the key, the horn button and the seat are rigid bodies on joints with return springs, detents and dampers; the truck reads only their joint angles, as its linkages and switches do: throttle, inching and brake fractions, spool fractions (12 % dead band), the selected direction (stalk detents), the parking brake (an over-centre lever), the key position (OFF, ON, START spring-returned), the horn switch (closes at 2 mm of button travel) and the seat switch (the seat's suspension deflection under the operator, filtered, closing at 180 N). No other code can set these signals. The pedals' strokes are measured to the toe board: brake 64.0 mm and inching 66.7 mm at the pad, accelerator 50 mm. The forces at the pad or knob:

Control	Force	Source
Accelerator	7.5 N to start it, 35 N at full travel	estimate (25-40 N class)

Control	Force	Source
Service brake	its return spring 10 N to start, 28 N at the floor stop, plus the line pressure's reaction: 450 N at 70 % of the travel, 1,800 N at the floor stop	spring estimate; anchors from ISO 6292 (section 3.4)
Inching pedal	14 N to start, 60 N at full travel by its own spring, plus the brake's force past 17 % of the travel	estimate; TCM's design
Lift, tilt, side-shift levers	1.7 N to leave centre, 15 N at full travel	estimate (10-20 N class)
Parking-brake lever	about 60 N to pull it over centre, 45 N to push it off	authored
Key	spring return from START, 0.3 N·m	estimate
Horn button	3 N preload, switch at 2 mm	estimate

The standard fixes only the brake's operating force (450 N); the other control forces remain class estimates (section 8), and the robot fixtures of section 10 report the force each control actually took.

4. The operator: the seated H1-2 and the driving-robot kit

4.1 The seated humanoid: model and seating

The seated operator is built from whichever humanoid rig is selected (`operator.py`): everything robot-specific comes from the catalogue model (its frames, its groups' roles, its `forklift` controller profile, its seated pose) and its URDF tree; the few values that must be measured per robot (belt wrap points on the pelvis, hull-artifact collision filters, the thumb's rest angle, where the hands rest, the feet) sit in one table, `SEAT_FIT`, keyed by model. The default operator is one floating Unitree H1-2 with Inspire hands (rig `forklift-h1-2`, controller profile `forklift`). It is seated by contact only, with no weld and no fixed base, held by its weight, the cushion's friction, its posture and a lap belt: two tension-only webbing halves with a dual-sensitivity emergency-locking retractor modelled on UN R16's type 4 (the class forklift seats such as the Grammer MSG 20 carry): a vehicle sensor, a pendulum on the seat's apparent gravity that locks when it tilts more than atan(0.45) = 24.2° (R16 6.2.5.3.1.1: locked by 0.45 g; no lock at 12° of tilt, lock above 27°; FMVSS 209 S4.3(j): locked by 0.7 g), so it answers braking and a tipping truck alike, and a webbing sensor on payout acceleration at 3.0 g (R16 6.2.5.3.2) with a 10 ms response; 7 N of retraction (R16 6.2.5.3.4). Forces act on the pelvis and, equal and opposite, on the seat. The robot's contact friction is in the ledger (`MU_ROBOT_SHELL` 1.0, Newton's default made explicit and open; `MU_ROBOT_PAD` for the finger pads); under MuJoCo's larger-of-the-pair rule the shell's value, not the cushion's 0.6, sets the robot's grip on the seat. The seat is a suspension seat with 80 mm of travel (Hyster J2.2-3.5XN), its spring and damper set for the operator's weight. The start posture leans the pelvis and trunk 5° forward on the hips (section 2.1) and is solved by IK on MuJoCo's own geometry (buttocks 3 mm above the cushion, back 4 mm from the backrest, feet flat on the footwell plate, hands on the thighs, every robot geom checked against everything it can touch) and settled by physics before tick 0.

The H1-2's thigh links and its pelvis/torso are kept from colliding with each other: their convex hulls (MuJoCo collides hulls of the link meshes) met with 1.36 kN when the right leg turned inboard toward the brake pedal, while the real meshes were 9.5 mm apart with no point of either inside the other (measured with exact winding numbers on the URDF meshes). Each palm and its thumb's proximal link are likewise kept apart: their hulls overlap in every thumb pose, and the resulting chatter would shake the truck through the seat. The legs still collide with each other, the seat and the cab; the hands collide with everything else.

4.2 Motors and actions

Each actuated joint runs the Unitree low-level command $\tau = \mathrm{kp}\ (\mathrm{q}^* - \mathrm{q}) + \mathrm{kd}\ (\mathrm{dq}^* - \mathrm{dq}) + \tau_\mathrm{ff}$, saturated by a torque–speed envelope (motoring $|\tau| \le \tau_\mathrm{max}\ (1 - |\mathrm{dq}|/\omega_\mathrm{nl})$, braking $|\tau| \le \tau_\mathrm{max}$; τ_max and ω_nl the URDF limits; the envelope shape is an estimate). The kd term is integrated implicitly. The Inspire hands' mimic followers are driven by the URDF's equalities and are not commandable. Reflected rotor inertia per joint class is a ledger entry (`ROBOT_ARMATURE`; the fingers' 0.01 kg·m^2 per finger is the middle of an open 0.003-0.03 range).

The model is weaker than the vendor's figures, and this is kept, not patched: the URDF's arm torque limits are 3-6 times below Unitree's published joint torques, and the Inspire finger joints' 0.5 rad/s against the vendor's about

260°/s (4.5 rad/s). The URDF is what the shared catalogue ships and what every robot environment here uses; raising it for this truck would make the forklift's H1-2 a different robot from the others'. Its effect on the controls is measured, not assumed: section 10 reports the torque each fixture needed against these limits, and slower fingers lengthen grasp and release (the key and the knob) but do not stop any control.

4.3 Which catalogue robots can drive this truck

Two operators are offered: the H1-2 (the default, `--robot h1-2`) and the driving-robot kit (`--robot drive-kit`, section 4.4). Every other setup is refused unless it is exactly a rig the operator builds (`robot_setup.py`), rather than swapped for one. The refusals below are the catalogue's own text, which `gar` returns with the refusal.

The G1 (rig `forklift-g1`, Dex3-1 hands) is refused **on a physical measurement**, not on a reach estimate. It seats and settles in the cab, and in E14b-style assisted fixtures (E14g: pelvis held at its seated pose, the working wrist or foot guided by a bounded force, the control moved only by contact) it works the lift lever with one finger (pulled 91.0 %, pushed 89.7 % of its travel, the forks up 0.44 m and down 0.63 m) and carries the spinner knob a full circle one way (the wheel turned 353°) and 10° the other way before the knob leaves its Dex3 hand (the wheel turned 50°). It cannot brake. Seated with its back on the backrest and the seat at its forward limit, its straight leg puts the toe on the service-brake pad only at rest (4.5 mm off), 30.9 mm short of it at half travel and 58.8 mm short at full travel, and in physics its foot never moved the pedal (0 % of travel). Slid 15 cm forward on the cushion, its foot pressed the pedal to 75 % of the travel (934 N on the pad) only while the external support held its pelvis there against the pedal's reaction (1.5 kN at the peak), its right ankle pressing the inching pedal too (312 N) and its left ankle the hood (251 N). A G1 could not stop the truck from a seat it can hold by itself, so it is not admitted. The H1-2 in the same G1 fixtures, as their positive control, holds the brake at 48 % of the travel and turns the knob 132°. The fixtures are therefore no proof that a robot which fails them cannot work the control; the H1-2's own fixtures (section 10, E14b) press the brake past the standard's 450 N.

The catalogue's arms (FR3, dual FR3, dual Panda) are refused because the shared builder welds an arm's base to the world frame, so it cannot ride in the moving cab, and because two hands leave no limb for the three pedals a driver works while steering and lifting: the service brake 10-1,800 N at its pad over 64 mm (the master cylinder's line pressure pushing back), the inching pedal 14-60 N over 67 mm by its own spring and the brake's force past 17 % of its travel, and the accelerator 7-35 N over 50 mm. Each refusal names the driving-robot kit as the non-humanoid route.

4.4 The driving-robot kit

The kit (model `drive_kit`, rig `forklift-drive-kit`, 105.3 kg) is the equipment vehicle testers bolt into a car or truck to drive it without a driver: one actuator per control, each modelled on a product class with that product's published limits. It is bolted to the chassis before the run: a floor frame carrying the IMU, the pedal unit, a lever bank above the cowl levers, the steering robot's stator, the key robot's mount, a bridge over the steering wheel for the horn robot, and a camera mast hanging from the overhead guard. A 67 kg steel ballast strapped to the cushion closes the operator-presence switch, as on driverless test vehicles (657 N on the seat in E14k).

Actuator	Product class	Published limit (at its output)	Contact radius
steering	AB Dynamics SR 15 Orbit, hollow, behind the rim; a fork round the spinner knob	20 N·m up to 200°/s, 15 N·m at 500°/s, 10 N·m at 1,000°/s	114 mm
accelerator	AB Dynamics Accelerator Robot (AR 1)	150 N up to 0.3 m/s, 130 mm stroke	124 mm
brake	AB Dynamics RBR 600 (the CBAR 600's actuator)	750 N up to 1.0 m/s, 500 N at 1.3 m/s	193 mm
inching	4activeDR brake-actuator class, in the place of AB Dynamics' clutch robot	350 N up to 0.4 m/s, 150 mm stroke	197 mm
lift, tilt, side-shift; direction; parking brake	automatic-gearshift class (GSA9001S figures, search); no hand-brake actuator product was found (open)	300 N up to 1 m/s	335 mm; 103 mm; 153 mm
key	a rotor coaxial with the key, two spring-closed jaws on its head	1.0 N·m, 3 rad/s, 30 N jaw preload (estimates: product classes only)	on the axis
horn	a rocker on the bridge, its tip over the horn button	30 N, 0.1 m/s (estimate)	231 mm

Only the end effectors collide, and each only with the control it works; housings, arms, mounts and the ballast carry mass and are drawn but have no colliders. Nothing of the kit is attached to a control: every force on a control is a contact force. Each actuator is one position servo about the axis of the control it works (`KitMotors`): τ = kp (q* − q) + kd (dq* − dq) + τ_ff, clipped to the product's force-speed envelope at its output, using the contact radius re-read every few ticks. A pedal's foot meets its pad after a small gap, so a pedal at fraction f of its travel needs q* = press_sign (f travel + gap); each pedal's sign, gap and travel are in the session's capabilities.

E14k-1 (SCRIPTED KIT, start-and-park seed 0, every actuator in turn): the key at START cranks the engine to running (617 rpm) and OFF stops it; the parking brake is released and set twice; the direction lever selects F, N and R; the accelerator goes to full travel at 150 N; each mast lever is pulled and pushed through its travel (spool ±1); the steering wheel follows the rotor within 1° over one turn left and two turns right; the horn switch closes. The brake robot at its published force (760 N along its path, the 2 % over 750 N being the one-step lag of the limit; 1,155 N net contact with the force across the path) takes the pedal to 78 % of its travel and the line to 14.4 MPa, against 9.45 MPa at the standard's 450 N, so it brakes harder than the standard asks but fails the 95 % travel criterion. The inching robot at 350 N takes its pedal to 87 % and fails the same criterion. 36 of 38 criteria pass. No end effector touches anything but its control. The negative control (every end effector's contacts switched off, the same commands) passes all 36 of its checks: no control moves and the engine never runs.

E14k-2 (SCRIPTED KIT: the autopilot's privileged state turned into the kit's own actuator commands, validated as a policy's; nothing else acts) completes start-and-park seed 0 in 27.05 s and seed 977 in 35.0 s, and aisle-cruise seed 0 in 26.95 s, each scored 1.00 by the unmodified evaluator. The steering robot sat at its 20 N·m limit for 0.07-0.11 % of the time, and no end effector touched anything but its control. These three runs are all the kit driving that was measured: the kit has not driven the other twelve tasks, and no film of the kit exists.

5. Dock, trailers, yard and loads

- **Trailer** (`dock.py`): one free rigid body (oak floor as a drivable surface and a collider, walls, nose, roof, rear frame with an ICC bar) carried by support elements computed every tick: the tandem axle group (air spring and damper, braked tyres with presliding stick and Coulomb slip, chocks as one-sided stops) and the landing gear (stiff legs, sand shoes with friction); an optional rotating-hook restraint at the rear impact guard; at the yard ramp, two stabilizing jacks under the nose (one-sided supports, touching with no preload at the start). Trailer creep is not written anywhere: it can only emerge when the truck's braking pushes the trailer harder than its tyres and shoes hold. The tandem's air springs are sized so the laden truck's front axle at the sill lowers it 4.25 in (`TRAILER_K`, **fitted**), the middle of Hendrickson L816's 3.5-5.0 in band for an air-ride trailer **without** a dump valve (trailer drop 2.5-3.5 in plus tyre deflection 1.0-1.5 in). A dump-valve trailer lowers 2.5-3.5 in before loading and then drops only by its tyres. The static load that sets the springs' rest lengths is the trailer's own weight, the loads standing in it and the leveler's or ramp lip's share, never a load on the truck's forks.
- **Leveler**: a hinged deck and lip resting on the trailer floor, deployed before the episode and floating with the trailer; deck and lip are drivable surfaces.
- **Yard ramp lip**: four 16 in steel plates, each pinned to the level-off's front edge and free between its 3 in down stop and upright, resting on the trailer floor by contact; each is a drivable surface. When the laden truck sinks the trailer more than 3 in, the plates hang at their stop and their toes stand above the floor: in the yard demonstrations the trailer's sill sank 101 mm under the laden truck going up (a 24 mm step at the toe, the plates on their stop for 3.8 s) and a further 23 mm coming down, the truck starting inside it (5.5 mm, 1.4 s; section 9.8), the physical limit of a 3 in lip on this trailer. A lip fixed to the world leaves a 70-128 mm step.
- **Stored rack loads**: wrapped unit loads on GMA pallets, 500-1,000 kg, 0.9-1.2 m tall so that every load clears the beam above it by the 6 in lift-off clearance (section 2.1), in 80 % of the locations. A scan of 1,000 seeds of every task finds no overlap and no clearance under 0.1524 m. Around the put-away slot each is one body (pallet and load) resting on the beams by contact and watched like the rack, so a carried load or the truck that strikes one is a strike and a load pushed off its beams is a failed put-away.
- **Conveyor** (`machines.py`): 2.5 in steel rollers on 0.10 m centres, each a body on a revolute joint with a velocity servo (a chain-driven gear motor); entry and end photo-eyes (MuJoCo rays) start and stop the drive.

- **Loads** (`loads.py`): GMA 48 × 40 in pallets (deck boards and stringers as box-shaped convex meshes, 20 kg) with stretch-wrapped loads, steel coils on chocks, or loose column-stacked cartons (5 layers of 6, 12–20 kg each); OCC bales (60 × 30 × 48 in, 450–590 kg) on 4 × 4 in dunnage strips. Friction pairs by material priority (forks over pallets 0.35, cartons 0.30, bales 0.40, wood 0.5; floors, racks and plates 0.50 so that the load's own coefficient sets the pair).
- **Every box** (floors, trailer and leveler parts, pallet boards, wrapped loads, bales, dunnage strips, cartons) collides as an 8-vertex convex mesh, not a MuJoCo box primitive: MuJoCo's box–box routine produced two 20 m-deep contacts between the 52 m floor box and an upside-down carton at rest on it, threw a pallet off a 16 m trailer floor box, gave two pallets one contact 1.22 m deep, and launched two 0.5 m cartons in a stack at 11 m/s. The fork blades' eight pieces are the only box primitives, and no box can meet them.

6. Interfaces: actions, observations, evaluation, CLI and HTTP

GAR id `forklift` (aliases `steer-forklift`, `forklift-truck`, `drive-forklift`, `lift-truck`), backend `newton_forklift`. Robot selection: `--robot h1-2` (the default) or `--robot drive-kit`. One action is a set of motor channels held for an exact number of 0.5 ms ticks (`named_joint_target`, 4 ticks to 30 s), with `.dq`, `.kp`, `.kd` and `.tau` suffixes on each channel; unnamed channels keep their last value; a command outside a motor's limits, with an unknown channel or of illegal length is refused before anything executes.

- **H1-2**: the Unitree low-level command per joint, `"h1_2/<joint>"` for q* in rad; the Inspire hands' mimic followers are not motors.
- **Driving-robot kit**: one channel per actuator, `"drive_kit/<actuator>"` (steer, accelerator, brake, inching, lift, tilt, shift, direction, park, key, horn). q* is the angle of the control the actuator works, so a pedal at fraction f of its travel needs q* = press_sign (f travel + gap), each pedal's values listed in the capabilities; the key robot's jaws are spring-closed, not motors.

Policy observation: each motor's position, velocity and torque, the IMU (roll, pitch, yaw and angular rates: on the H1-2's pelvis, on the kit's frame), and the policy camera's image (the H1-2's head camera or the kit's cab camera) with its frame stamp. Owner key adds the full state, the evaluator, checkpoints and the diagnostic cameras. Task selection: `gar run forklift --select task=<task> --select seed=<0-999>`; seeds 0–799 train, 800–899 validation, 900–999 test. Checked live: a `gar run` session with `--policy-key` committed a policy-key action through HTTP, served the policy key its observation and head-camera image (the operator's view over the wheel, Figure 17) and refused it the full state and the evaluation (403), while the owner key got both and the session dashboard. That check ran on an earlier build of the world. The served runtime measurement of section 13 ran 46 `gar run` sessions across both robots, with both keys, and timed 3,153 requests, every one answered 200.

What a policy and its operator can rely on:

- **Capabilities state the costs, per robot.** The backend's `hardware` block carries the session's own robot's measurements (section 13): the shortest command none of whose served requests exceeded 200 wall s per simulated s (`shortest_command_within_200x_s`: 20 ms for the H1-2, 0.1 s for the kit), the wall seconds per simulated second per request by command length, what a new camera frame costs, the render worker's GPU memory and the session's host memory, the trace's cost, the longest command (`max_command_s`, 30 s), the client timeout (`client_timeout_s` 1,800 s: a 30 s command at the slowest traced physics plus a cold render worker, with margin for a busier host), and the render worker's release interval and cold-start cost. The worker stays warm 600 s after the last request, so a policy that pauses to think does not pay a 71-99 s cold start on its next observation.
- **Refusals before anything runs.** A non-integer or out-of-range seed (`--select seed=1.5`, strings, booleans) is refused, and so is any robot setup that is not exactly a rig the operator builds; the backend checks the rule again before building and confirms after building that the operator built that rig.
- **Every tick is traced.** The trace route records every tick of every command: truck pose and speed, the controls' signals, drivetrain, steering and hydraulics telemetry, mast joints, tyre loads, every load's pose and velocity and the operator, with the events (strikes, wheel lift, overturn, a load touching the floor, horn, intersection entries, seat switch, faults). It costs 0.17-0.59 ms of wall time per 0.5 ms tick, 3.5-18.5 % of a

traced command's time (section 13), keeps the last 44-96 simulated seconds in its 256 MiB budget, and can be turned off (`renderer_options trace=false`).

- **Installable and checkable.** `forklift.visuals`, the asset source lists and the robot installers are packaged. The asset lock covers all 2,290 files (2.53 GB, every one hash-checked; a fresh install downloads and verifies them). `gar doctor forklift` checks the locked files' presence and sizes, sample hashes, the H1-2 files, that every offered robot's builder (the H1-2's and the kit's) imports, and the renderer; a failed row names the command that fixes it.

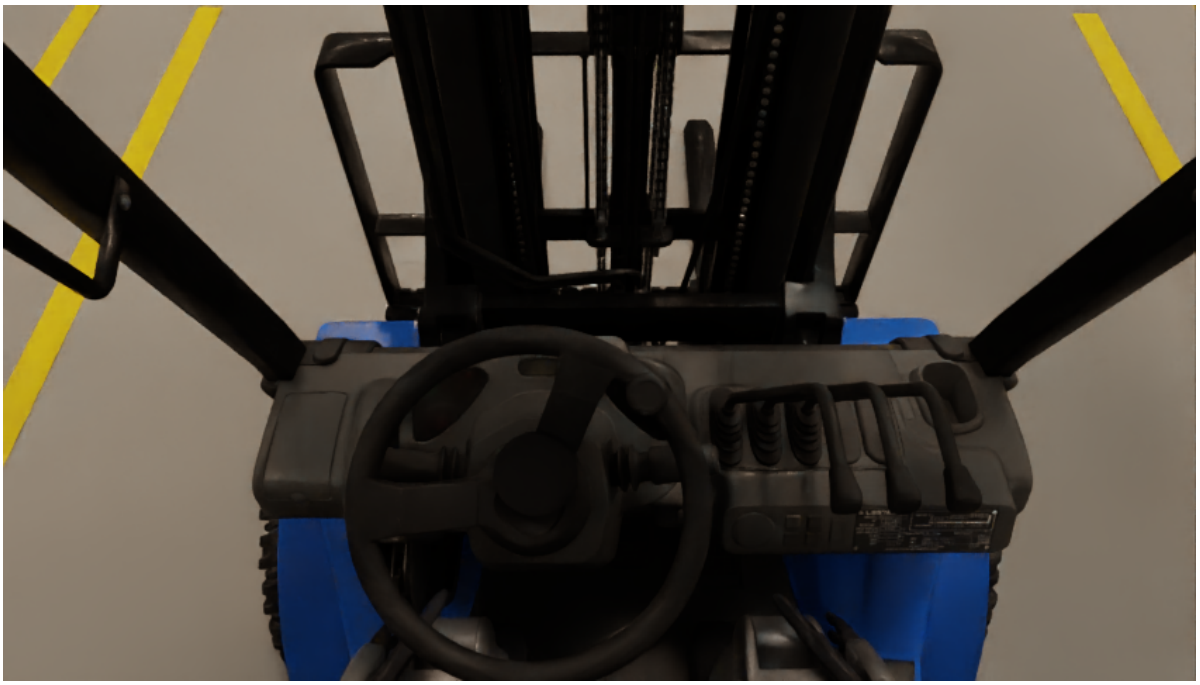


Figure 17. The head camera image a policy key receives over HTTP at the start of start-and-park (an earlier build).

7. Tasks and the evaluator

Fourteen tasks in three presets (`task.py`; catalogue `benchmark.json`; `data/variations.json` freezes every seed's resolved start). A seed (0-999) sets the task's target and the start state. The target varies by task: staging lane S1-S4 (drive-to-pallet); one of ten floor bays (carry-to-floor-bay, carry-loose-stack); a rack location (put-away-rack: any of the four rack faces along the two working aisles, bays 1-7, the first beam level in two seeds of three and the second in the third, either pallet position); dock door 1-4 and the trailer side (load-trailer, load-trailer-chocked, unload-trailer, whose staging lane is its door's); door 3 or 4 (load-bales); the direction, east or west (aisle-cruise); the start (start-and-park: either end of the main aisle or lane 1; feed-conveyor: lane 2 or the receiving lanes R1-R3); the trailer side (ramp-up-loaded); the drop zone east or west of the ramp (ramp-down-loaded); the route (yard-route: from the apron to the ramp foot, or from the ramp foot to the apron line). The start state varies too: truck pose; load masses and heights (unit loads 600-1,200 kg and 0.9-1.5 m on the floor-bay task, 600-1,000 kg and 0.9-1.2 m for put-away, 600-1,100 kg and 1.0-1.4 m for load-trailer, 900-1,300 kg and 1.0-1.3 m in the yard; the trailer's nose pallets 500-900 kg); the target pallet's yaw ±10° in drive-to-pallet; trailer bed height 44-52 in; wet patches in a third of the warehouse seeds and 40 % of the yard-route seeds; and in one load-trailer seed of five the trailer arrives unsecured. Each instruction names its target. Time limits are about three times an expert's time.

Task	Preset	Instruction (abridged)	Difficulty	Limit
start-and-park	warehouse	start, release the brake, drive to the parking bay, park inside its lines: forks lowered and tilted to the floor, neutral, brake set, engine and key off	easy	90 s
aisle-cruise	warehouse	drive the main aisle in the named direction at the posted 8 km/h, slow and sound the horn at both intersections, stop at the far line	easy	120 s
drive-to-pallet	warehouse	stop square in front of the pallet in the named staging lane, forks level at its openings, just short of it	medium	120 s

Task	Preset	Instruction (abridged)	Difficulty	Limit
carry-to-floor-bay	warehouse	set the pallet on the forks down inside the named floor bay, back out, lower the forks	medium	180 s
carry-loose-stack	warehouse	the same with a pallet of loose cartons, spilling none	very hard	240 s
put-away-rack	warehouse	put the pallet away in the named rack location (aisle, bay, beam level and height, left or right position), back out, lower	hard	240 s
feed-conveyor	warehouse	set the 2 t coil pallet on the press infeed conveyor between its rails so it is conveyed to the press, back away, lower	hard	180 s
load-trailer	dock	load the pallet into the trailer at the named door and side, directly behind the nose pallets; back out; on a red restraint signal, wait at the door short of the leveler instead	hard	240 s
load-trailer-chocked	dock	the same at a door with no restraint, the trailer held by its spring brakes and wheel chocks	hard	240 s
load-bales	dock	load two OCC bales from the bale row into the trailer at the named door, side by side on the dunnage strips at the nose, against the front wall; back out onto the dock	very hard	480 s
unload-trailer	dock	unload the named nose pallet, set it down in the door's staging lane, back out, lower	very hard	300 s
ramp-up-loaded	yard	drive up the yard ramp into the trailer with the pallet and set it down at the nose, against the named side wall	hard	240 s
ramp-down-loaded	yard	bring the pallet out of the trailer, down the ramp, set it down in the named drop zone, back out and lower	hard	240 s
yard-route	yard	drive the marked lane to the ramp foot, or back to the apron, and stop square at the yellow line	easy	150 s

With the spring brakes set the chocks carry no force in this model (section 9.8), so load-trailer-chocked is load-trailer at a door with no restraint, not a harder trailer.

Shared criteria (every task): no strike of the truck, its load or the robot on a watched structure above 2,000 N or 50 N·s (racks and the loads stored in them, columns, walls, conveyor guides, trailer walls, bumpers, curbs, parked trailers, dressing within reach). A load's contact is exempt only in three cases. (1) A gentle set-down on rack beams: the part it bears on at the peak is a beam, its onset $\leq$ 0.10 m/s and its per-contact peak $\leq$ 25 kN (measured: 12.9 kN for 865 kg set down at 0.045 m/s and 53 kN at 0.29 m/s, interpolated to 0.10 m/s and scaled to the heaviest 1,020 kg unit). (2) A gentle set-down on a trailer floor or the conveyor's rollers: onset $\leq$ 0.10 m/s along the contact normal relative to the surface, and a peak within m (g + B v), the most force the contact law gives a load of mass m arriving at that speed (79.6 kN on a trailer floor for the heaviest 1,320 kg unit, 121.8 kN on the rollers for the 2,020 kg coil pallet). (3) A load resting or leaning against a structure: onset under 0.10 m/s and never pressed harder than its own weight (2 kN at least). So a slam onto the beams at 0.43 m/s (109 kN) or a pallet dragged out with the forks is not exempt. Also shared: no wheel off the ground for more than 0.3 s; roll under 6° and no overturn; no carried load touching the floor except at its destination (a floor contact is recorded only after the load has been off the floor for 0.25 s, so a pallet resting where it started is not a drop); never above the posted 8 km/h + 10 %, inside buildings and in the yard; the seat switch never open more than 1 s while moving; no contact of the robot with the overhead guard above 130 N (ISO/TS 15066's skull and forehead limit, `head_clear_of_guard`); the time limit.

Task criteria (`evaluator.py`), each with its measured value, threshold and source in the evaluation record:

- start-and-park: the whole footprint inside the parking bay's lines; at rest $\geq$ 2 s; parking brake applied, direction neutral, engine off, key off; fork tips $\leq$ 20 mm above the floor, and tilted to the floor: both tips $\leq$ 5 mm and the blades not pitched back (OSHA (m)(5)(i): forks fully lowered).
- aisle-cruise: in each of three straight sections $\geq$ 80 % of the distance at 6.5–8.0 km/h; each intersection entered at $\leq$ 5 km/h with the horn sounded in the 3 s before (OSHA (n)(4)); the truck centre within 0.6 m of the aisle centreline; stopped with the front wheels within ±0.2 m of the far line.
- drive-to-pallet: at rest $\geq$ 2 s; fork tips 0.20–0.30 m short of the stringer ends (OSHA eTool 8–12 in); blades level within ±1°; blade bottoms 20–60 mm above the floor (the GMA opening); each blade $\geq$ 40 mm from the stringers and the fork centre within ±50 mm of the pallet centre; the fork axis within 3° of the pallet axis.
- floor-bay tasks: all four pallet corners inside the bay lines (90 mm margins), yaw within 5° of the lines, corners on the floor within 10 mm, fork tips $\geq$ 0.5 m from the pallet, forks at or below travel height (0.15 m), at rest $\geq$ 1 s; loose stack: no carton moved more than 30 mm on the pallet and none ever touched the floor.

- put-away-rack: the pallet on both beams of the instructed location's own frame, bay and level (its underside within 30 mm of the beam top), front overhang 51–101 mm past the front beam (the rack-loading practice range), $\geq$ 50 mm to the upright and to a neighbouring pallet, level within 2°; the fork tips more than 1 m from the pallet and at most 0.30 m above the floor; every stored load (the ones around the slot are bodies) within 30 mm of where it stood.
- feed-conveyor: the pallet inside the conveyor frame, square to the rollers within 3°, the end photo-eye registered it, the forks withdrawn and at or below travel height (0.15 m).
- load-trailer with a red restraint signal (one seed in five): the truck's loaded front never past the leveler's inner edge (OSHA 1910.178(m)(7)); waiting at the door, its loaded front past the end of the staging lane's lines; at rest $\geq$ 2 s with the parking brake set and the direction in neutral; the load still on the forks.
- load-trailer (both, secured trailer): inside the trailer resting on its floor; $\leq$ 50 mm behind the nose pallets; $\leq$ 125 mm from the named side wall (derived: at full side-shift the 40 in pallet's edge is 0.608 m from the truck's centreline and the asset's rear tyres 0.682 m, so the pallet cannot come closer than 74 mm plus the tyre's clearance, and a 50 mm threshold no operator of this 1.36 m-wide truck could meet); level within 2°; the truck back inside the building; the trailer moved $\leq$ 25 mm from its spot.
- load-bales: exactly two bales in the trailer, each upright within 5° and resting on dunnage strips (its bottom within 30 mm of the strips' tops and its centre over the strips it bears on, read from the strips' current poses), its front face $\leq$ 75 mm from the front wall at both of its front edges (a bale touching the wall at one corner and turned away from it does not count); the two not overlapping ($\leq$ 5 mm, on oriented footprints), at any spacing, either bale on either side; the whole truck, fork tips included, behind the door face; the trailer moved $\leq$ 25 mm. A bale's or a dunnage strip's contact with the target trailer's nose wall is a strike only if it approached the wall at $\geq$ 0.10 m/s or pressed on it harder than its own weight (2 kN at least), and then only above 2 kN; its side walls keep the shared rule, so a bale lowered onto its strips while touching the wall and resting there at 300 N is not a strike. Its score is a progress reward, not the fraction of criteria met: 1 on success, 0 for a faulted world, else 0.9 × progress × workmanship (`evaluator.BALE_REWARD`, `BALE_WORKMANSHIP`). Progress reads each bale's current state only (placed 1; resting on strips in the trailer short of the wall or tilted past 5°, graded down from 1; elsewhere 0-0.8 by its remaining distance to the nose row against its distance at the start; 0 while it rests on a floor or lies over), the mean of the two best bales, with the truck's exit counted in proportion as that mean rises from 0.9 to 1. Workmanship is the product of a factor per broken site rule, graded by how far it was broken: a strike per contact event (episodes less than 0.25 s apart are one event), by its approach speed's kinetic-energy ratio $(\text{v} / 0.10\ \text{m/s})^2$ or, for a press, its peak force over 2 kN, never by a resting contact's accumulated impulse; a bale's floor contact per contact event (latched entries less than 0.3 s apart are one), 0.8 to 0.9 by duration if it came off the floor again, 0.8 if it lies on a floor, each times a factor falling with the contact's approach speed above 0.10 m/s (recorded at each new floor contact), the same on every floor, a trailer-floor arrival charged once. Doing nothing scores 0 (0.909 under the criteria fraction); every number in the reward is authored. The strike record (`events.py`) also latches, per structure part, the fastest approach among episodes above 2 kN, so a nose-wall strike cannot drop out of it. The dock posts its 8 km/h limit on its east wall (`visuals/dockyard.py`, `sign_speed_dock`). Strips in a van depart from published mill practice. Weyerhaeuser, "Introduction to Shipping" (DTP 2024, 02/2004), §4.03 Van Truck Loading: "Do not load with pallets or stickers on vans." and "A bale clamp, not straight forks, is used to unload."; §4.06: "Trailers are unloaded with bale clamps, not straight forks." (https://www.pfri.uniri.hr/bopri/documents/Unit1d-IntroductiontoShipping-TRUCKS_000.pdf). Evergreen Fiber, "Safe truck/trailer loading procedure": "Do not load with pallets." (http://www.evergreen-fiber.com/files/van-safe-loading.pdf). The strips are there because this environment's bale is a rigid flat-bottomed box: straight blades under it carry its whole weight and cannot be withdrawn unless something else bears it. Real alternatives are a bale clamp, a different attachment, or a bale whose formed bottom leaves the blades clearance (LoadKing, US 6,474,226 B1: "a desired arched bottom surface 82 of the bale 36 allowing fork elements 84 of the lift truck, by way of example, to easily slide under the bale"), which changes the bale. Neither is modelled here.
- unload-trailer: the pallet inside the door's staging lane on the floor (within 15 mm); the other nose pallet moved $\leq$ 30 mm; fork tips $\geq$ 0.5 m from the pallet and at or below travel height; at rest $\geq$ 1 s.
- ramp-up-loaded: the pallet $\leq$ 75 mm from the trailer's front wall on its floor and $\leq$ 125 mm from the named side wall (as load-trailer); roll-back $\leq$ 0.3 m (travel against the selected direction on the ramp; a deliberate reverse is not roll-back).
- ramp-down-loaded: the pallet inside the drop zone, upright, its corners within 10 mm of the ground; $\leq$ 6

km/h on the ramp (OSHA (n)(7): "slowly"); fork tips ≥ 0.5 m from the pallet and at or below travel height; at rest ≥ 1 s.

- yard-route: the footprint never crosses the lane's outer lines; fork tips within ± 0.3 m of the ramp-foot line (or, going back, the apron line at the lane's head), yaw within 3°, at rest.

The evaluator reads simulated state only (poses, contacts, signals, latched events); it never reads the robot's commands. Its geometry is fixed by the task and seed (`scene.zones`).

8. Parameter ledger

Every physical constant a model reads is one entry in `params.py` with its unit, provenance label and source. The ledger holds 252 entries: 97 estimates, 31 authored, 29 derived, 28 read from the primary source, 17 measured on the asset or the model (one of them on a manual's drawing), 9 from searches, 8 from the repository's earlier environments, 7 open, 6 fitted, 1 an engine default, and 19 with mixed labels (for example read/estimate: a datasheet range whose value inside it is an estimate; read* marks a source read through a fetch summariser, not opened). A consumer check covers both directions: every entry is read by code, and every module-level number in the physics modules comes from the ledger or is listed there with its reason (units, contact sampling, regularisation, asset coordinates measured from the meshes, the seat fit measured per robot, collision bitmasks).

Fitted is a label of its own: a value set so that the model reproduces a read observable (the drive ratio and converter stall ratio to the drawbar pull and top speed; the pump displacement to the laden lift speed; the lowering regulator's setting and opening, and the lowering cap, to the sheet's lowering speeds; the lift leakage to a 35 mm creep; the trailer's springs to the sill drop). The probe that reproduces that observable checks the fit, so each probe reports it under `fit_checks`, apart from its verdict, and section 9 does not count it as a validation. The service brake is not fitted: its factor is derived from the braking standard and the TCM hardware (section 3.4), so E8's laden stop at the standard's force checks that derivation, not the world.

What carries the physics and how it is anchored:

- **Mass and geometry**: the H2.5A sheet (service weight, axle loads laden and unladen, load centre, speeds) and the asset's measured geometry (section 3.1 says which value comes from which vehicle). Individual body masses are estimates distributed to reproduce the sheet's axle loads and the derived CG; the CG height (0.68 m) is an estimate (open range 0.55-0.80 m) and is the quantity the stability experiments are most sensitive to.
- **Tyres**: vertical stiffness and damping fitted to a measured 18x7-8 super-elastic tyre (Oh et al. 2013) and scaled by width $\times \sqrt{\text{radius}}$; the SUPREM lateral law with its published coefficients and its drum's measured friction level on dry concrete; the longitudinal peak, slip scale and sliding ratio from one super-elastic tyre's braking-slip curve on the same drum (Pross & Bruns 2019) (section 3.3). The truck's pitch damping comes out of the tyres and is compared with the one measured forklift value found (Haritos et al. 2013, ζ 0.031): E5b measures the laden pitch mode at 0.83 Hz with ζ 0.024-0.031.
- **Drivetrain**: the engine curve's anchor points are read; the converter stall ratio and the reduction are fitted; the laden top speed, the 0-15 m times and the gradeability are held out (section 9.1).
- **Steering**: displacement, relief, lock-to-lock turns, the manual torque limit and the neutral spring torques are read (from electric trucks' and Danfoss documents); the leakage is derived from a maker's allowed slip at the stops (Eaton); the kingpin axes are measured on the asset; priority flow and the linkage's stiffness are estimates.
- **Hydraulics**: relief and the sheet's lift and lowering speeds are read; pump displacement, the lowering regulator and leakage are fitted; cylinder bores derived; the stage stroke read from Hyster's mast table; the lift lines, mast friction, valve area and the end cushions estimates; the regulator's type and lag open.
- **Brakes and controls**: the service brake's hardware and pedal geometry are read from the TCM manual, its pressure-volume curve is anchored to the standard's pedal forces, and its factor is derived from the standard's drawbar drag; the parking brake, clutch capacity, pedal springs and seat-switch threshold are estimates or open, each sized from a stated physical requirement. The pedal strokes are measured on the asset; the other control forces are class estimates.

- **Facility and loads**: GMA pallet, rack, trailer, leveler, yard ramp (Bluff MobilePRO manual) and bale dimensions are read; the rack lift-off clearance is read; the aisle width is authored over the VDI 2198 formula; frictions are estimates or search-level; stored-load masses and dunnage placement are authored.
- **Operator and seat**: the guard height is the sheet's, the head-contact limit is read from ISO/TS 15066's table (through a reproduction), the seat's travel from a Hyster guide and the belt's locking thresholds from UN R16; the seat's spring and damper and the start lean are estimates or authored.
- **Driving-robot kit**: each actuator's force-speed limits are its product's published figures (read or search) in the kit's catalogue model, the key and horn robots' are estimates, and the kit's mounts and ballast are authored (`DRIVE_KIT_MOUNTS`, `DRIVE_KIT_BALLAST`).
- **Numerical**: 0.5 ms step, MuJoCo's Newton solver with 60 iterations and 50 line-search iterations, 10 no-slip iterations, impedance ratio 10, 4 ms rigid contacts, convex-collision tolerance 1e-8 (at MuJoCo's default 1e-6 resting pallets drop out of contact for single ticks, section 12.1).

The open values and the estimates with the widest ranges are listed again in section 15 with what each would change.

9. Component validation (physics experiments)

Each experiment states its prediction and its rejection rule before it runs, records its assistance, and writes a verdict (`passed`) that applies the stated rule; failed verdicts are kept and discussed. Each experiment records the full series with a plot, and every number below is read from those records. The truck is driven by phantom forces at its own controls (ASSISTED: forces at the pedal pads, lever knobs and spinner knob, reacted on the chassis like a mounted actuator), a 67 kg ballast on the seat standing in for the operator; nothing pushes the chassis.

Experiment	Prediction (written before running)	Measured	Verdict
E3 drivetrain, unladen	0-15 m 4.5 s ± 10 %, top 18.7 km/h ± 5 % (sheet; the top speed is **fitted**)	4.45 s, 18.72 km/h	pass
E3 drivetrain, laden	5.9 s ± 10 %, 18.4 km/h ± 5 %	5.15 s, 18.45 km/h	**fail** (13 % too brisk; held out, not refitted)
E3 idle creep, reversal	creeps at idle; reversal through clutch slip under its cap	4.9 / 4.1 km/h; 493 N·m of 600	pass
E4 steering	~4.8 turns, both stops at the relief, slip only by leakage there, a hand pump with the engine off, radius within 5 % of the derived 2.387 m	4.90 turns, 12 MPa, 3.5 rpm at the stops; engine off 5.36 turns within 120 N·m; 2.365 m	pass
E4r released wheel	moves $< 1°$ in 4 s after release	0.0014° and 0.0013°	pass
E5 mast hydraulics	sheet speeds ± 10 %, staging, relief caps the pump; a smooth rated-load cycle keeps the steer axle down (no rear-off over 0.3 s)	0.66 m/s unladen lift; staging; 18.5 MPa; rear-off 0.14 s at most	pass (abrupt cycles lift it longer, reported)
E6 tilt table, 3 cases	tips at the support triangle's angle (21.9 / 11.5 / 5.6°)	unloads at 20.3 / 9.8 / 5.1°, overturns 29.3 / 10.6 / 6.0°	pass (ISO 3.4° / 3.4° / 2.3°)
E7 steady turns	a wheel lifts near 3.6 m/s^2	2.5 and 1.7 m/s^2 at most, no lift	inconclusive
E7 run-ups	a wheel lifts entering a turn at 17 km/h	lifts at 13.1 km/h (30°), 15.4 km/h (20°)	pass
E8 service brake	dry, laden: 0.25-0.30 g at 450 N (by the brake factor's derivation); harder push brakes harder; unladen limited by the front tyres; energy booked within 10 %	laden plateau 0.167 / 0.252 / 0.276 / 0.370 g at 300 / 450 / 492 / 600 N; energy within 0.7 % (6.6 % at 600 N)	pass (the 450 N plateau checks the derivation)
E8 parking brake	holds 15 % laden, < 1 mm in 120 s	5.7 μm	pass
E9 pallet on the forks	slide onset inside the measured bracket, 3 tilts	3 of 3	pass
E9 loose stack	top cartons slide from ~2.9 m/s^2	none moved at a 2.78 m/s^2 peak; at 5.48 m/s^2 10 moved more than 30 mm, none fell	**fail**
E10 strikes	stopped by contact, ≤ 5 mm, strike latched	≤ 0.06 mm, latched at 1, 2, 4 km/h	pass

Experiment	Prediction (written before running)	Measured	Verdict
E11 dock	sag 89-127 mm; restraint $\leq$ 5 mm < chocks < none	115 mm; 3.0 mm; chocks = none = 8.9 mm (the chocks carry no force); the carried pallet struck the trailer floor in every run	**fail** (chocks < none refuted; strikes)
E11 trailer walk	chocked: the trailer walks; restraint $\leq$ 5 mm	5 entries each at 9.7 km/h: chocked 3.3-3.5 mm after every stop (12.3 mm at most during one), no accumulation; restraint 0.1 mm (3.0 at most)	**fail** (no walk)
E12 grades	rolls back on 4 % and 13.3 % at idle; hill start; coast within 5 %	2.11 m back on 13.3 %, none on 4 %; climbs; coast within 0.11 %	pass (the 4 % idle-hold prediction refuted)
E12 gradeability	4.8 km/h $\pm$ 20 % on 28 % laden, 26 % unladen (sheet, held out)	0.84 km/h laden; 6.69 km/h unladen	**fail** (both ways)
E12 yard lip	the plates rest on the trailer floor; a step only below 3 in of sag	24 mm step going up, 5.5 mm down	pass
E18 convergence	< 5 % from halving the tick	0.34 % / 0.21 % / 0	pass
E19 soak	20-50 mm fork drop, steady; truck, load, pelvis still	34.2 mm at 3.42 / 3.42 mm/min; $\leq$ 0.2 mm	pass
E20 doing nothing	no task succeeds by holding the start command for 3 s or 10 s	none of 45 task-seed runs	as predicted
E21 kingpin and straight line	the asset's kingpin boss is vertical at the pivot; hands off, the truck stays in its aisle	0.018°, 0.002 mm; 0.56 m sideways at most in 60 s (1.589 m allowed)	pass

The E8 rows are stops at a prescribed force on the brake pad. Other "full" pedal presses in these experiments take the pedal to its stop.

9.1 Drivetrain (E3) and gradeability (E12)

Phantom accelerator and direction lever, the truck on level dry concrete. Unladen it covers 15 m in 4.45 s (sheet 4.5 s $\pm$ 10 %: pass) and tops out at 18.72 km/h (the sheet's 18.7, a **fitted** value: the reduction is fitted to it, so it checks the fit). Laden (the rated load) it tops out at 18.45 km/h (sheet 18.4 $\pm$ 5 %: pass, held out) but covers 15 m in 5.15 s against the sheet's 5.9 s: outside the band, a held-out miss (13 % too brisk) that was not refitted. At idle in gear the stalled converter creeps the truck at 4.9 km/h unladen and 4.1 km/h laden on level floor (it must be held on the brake, as on a real torque-converter truck). Throwing the direction lever to reverse at 10.2 km/h, the clutch slips at up to 493 N·m (its cap is 600) and the truck ends backing at 12.8 km/h, with no torque spike.

Gradeability, held out: the sheet gives 28 % laden and 26 % unladen at 4.8 km/h. With the accelerator floored, the laden truck climbs the 28 % grade at a steady 0.84 km/h, the engine pulled down to 1,784 rpm by the converter near stall. Unladen, reversing up the 26 % grade (forks downhill), it climbs at 6.69 km/h with the drive tyres' slip at most 0.13, and the same with a foot that keeps the tyres below sliding. The laden truck is far too slow and the unladen one 39 % too fast, so both fail the sheet's speed. The sheet's figures do not give an independent check: its 28 % is the arcsine of its drawbar pull over the laden weight to four digits, so the 4.8 km/h asks for the drawbar pull at three times the speed at which the sheet states it, which a converter whose torque ratio falls with speed cannot give. E21b brackets the tyres' friction (its level scaled to 0.67, and a steeper load law): laden 0.81-0.83 km/h and unladen 6.38-6.74 km/h, so neither miss comes from the friction. The misses are reported, not tuned.

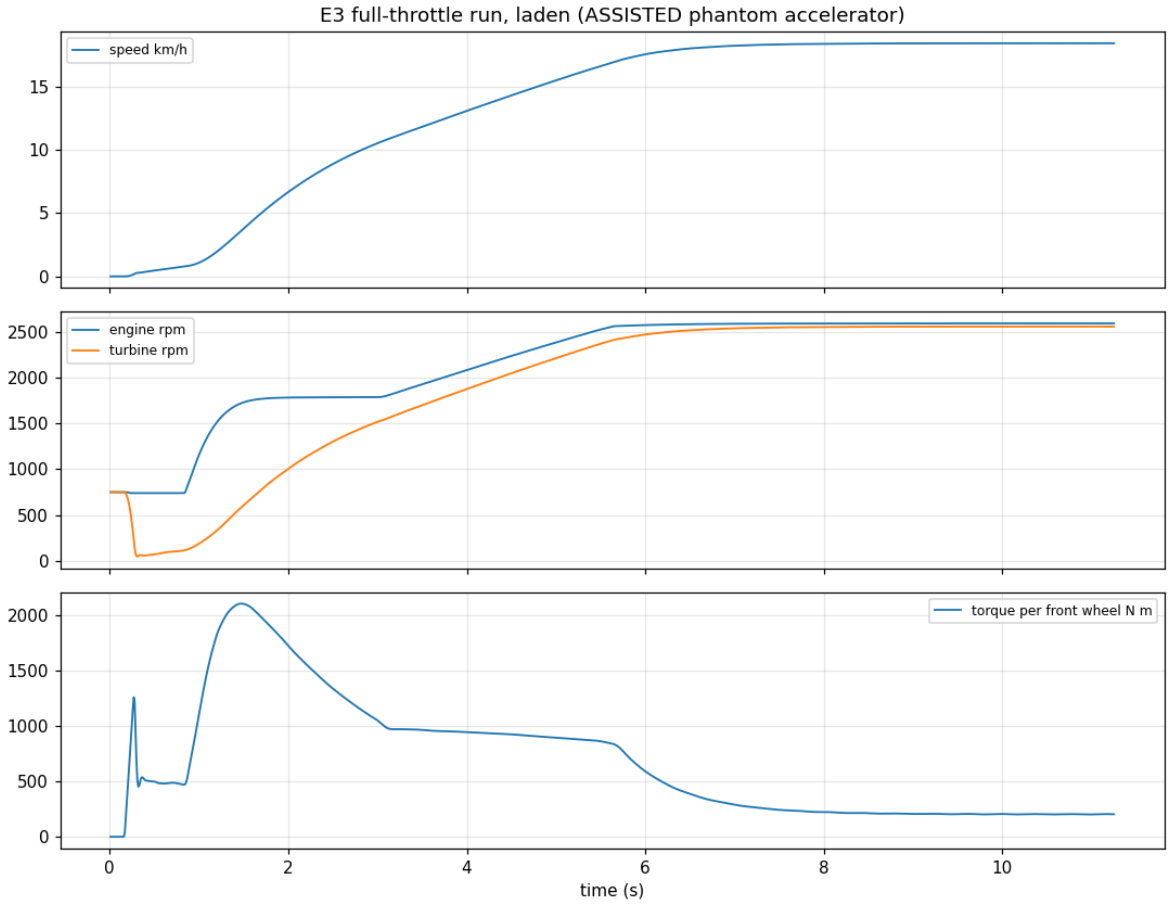


Figure 4. E3, laden: speed, engine and turbine speed, and torque per front wheel through a full-throttle run.

9.2 Steering (E4) and the kingpins (E21)

Phantom knob torque. Engine running: 4.90 turns lock to lock (the 4.8 that sets the cylinder's volume, within 10 %), up to 178 rpm at the wheel, both stops reached with the steering relief at 12 MPa, and against the stops the wheel slips 3.5 rpm by the unit's leakage, as derived. Engine off the unit is a hand pump: lock to lock in 5.36 turns at up to 8.7 MPa within the 120 N·m an average operator can exert (Danfoss). With the truck standing, 15 N·m at the knob moves the knuckles 0.80° in 3 s, 60 N·m 6.6°, and the 120 N·m cap 10.6° (72 N·m applied); the knuckles' travel is within 1.4 % of the oil the wheel pumped, and the hand's torque 1.14 times the cylinder's. The full-lock outside turning radius (counterweight) is 2.365 m against the 2.387 m derived from the asset's geometry (within 5 %; the H2.5A sheet's 2.263 m and the PrimeMatch's 2.357 m are for narrower trucks, section 3.1). Let go at −130° and at +90°, the wheel moves 0.0014° and 0.0013° in the next 4 s (E4r): a non-reaction unit holds it. All checks pass.

The kingpin boss of the asset's knuckle mesh is vertical within 0.018° and 0.002 mm from the authored pivot (E21a). Hands off for 60 s at 6 and 12 km/h, unladen and laden, the truck strays at most 0.56 m sideways (its aisle allows 1.589 m) and the knuckles move under 0.004°; with the leak at the 5 rpm end of its range, 0.69 m (E21): pass.

9.3 Mast hydraulics (E5)

Phantom lift lever, accelerator floored (the engine governs at 2,568-2,593 rpm). Unladen lift 0.66 m/s mean against the sheet's 0.65 (held out, within 10 %: pass); laden lift 0.64 m/s against 0.60, laden lowering 0.55 m/s against 0.58 and unladen lowering 0.48 m/s against 0.50 (all **fitted**: the pump displacement and the lowering regulator are set from them; all within 10 %). Lift speed scales with pump speed: at idle the laden lift is 0.17 m/s with the

engine pulled down to 717 rpm by the pump's torque. The free-lift stage rises first and the main stage lowers first, as the staged circuit should; the pump's pressure stays at the 18.5 MPa relief while raising (19.0 MPa peak on the closed lift column, whose pressure is its load's, and 19.8 MPa while the rated load is lowered). All verdicts pass.

With the closed lift columns, the lines' losses, the measured tyre damping, the stage-end cushions and the pump's load on the engine (sections 3.3-3.6), and the rated load at the rated moment (section 9.4), a **smooth** rated-load cycle to 4.44 m (lever opened over 1 s, closed over 1 s below the top, lowering lever opened over 2 s) lets the steer axle leave the floor three times, for 0.06, 0.14 and 0.12 s, inside the evaluator's 0.3 s wheel-lift rule (7.1 kN mean on the steer axle at height; the truck pitches 0.3-1.0° forward). **Abrupt** cycles (the lever fully opened and closed within 0.2 s) lift it longer: with the rated load at full throttle the steer axle leaves the floor six times, for up to 0.27 s, and once at idle for 0.36 s, past the rule. Stopping 2.8 t moving at 0.6 m/s within 0.2 s passes about 1.7 kN·s through the mast, which rocks a truck whose steer axle carries 7-8 kN; that is physics, reported, not judged, and it is why operators are taught to feather the levers. The abrupt cycles pitch the truck at most 1.6° forward and none overturns it.

E5b brackets the lowering regulator and the tyres' damping. With the load-compensated regulator, lowering the rated load keeps the steer axle loaded at every compensator lag tried (lowest 2,571, 2,571 and 462 N at 0, 0.05 and 0.15 s), and at 1,990 N with the tyres' damping at 0.7 of its value; a restrictor regulator instead empties the steer axle for 0.22-0.35 s at every lag. The laden truck's pitch mode at 4.44 m rings at 0.83 Hz with 0.024-0.031 of critical damping, around the one measured forklift value found (0.031, Haritos et al. 2013).

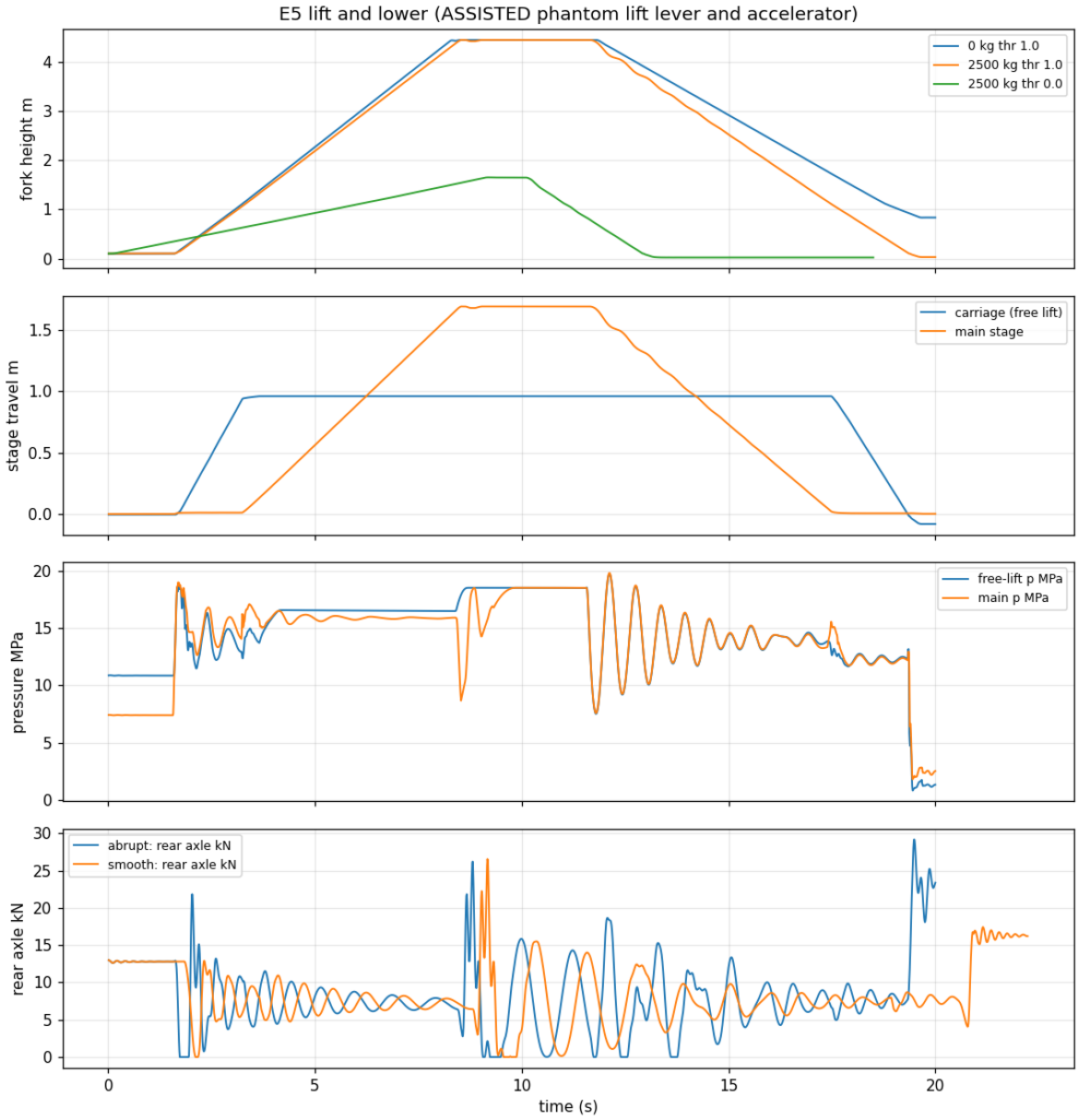


Figure 5. E5: fork height, stage travel, cylinder pressures and the steer-axle load through the lift cycles.

9.4 Stability (E6, E7)

Nothing in the code decides when the truck tips: it must come out of the mass distribution, the three-point support (two front tyres and the steer-axle pivot) and the tyres. E6 rotates gravity slowly (0.2°/s) about the parked truck, which loads it exactly as a tilting table does; the prediction is the support triangle's tipping angle atan(d / h) from the truck's own centre of gravity summed from the model's bodies. The laden cases use ISO 22915-2's mast attitudes (vertical for the longitudinal stacking test, fully back for the lateral one) with the rated load at 3.0 m. E7 drives the unladen truck in a circle at a fixed lock with the speed rising, and runs it up straight at 17 km/h before turning the wheel to a fixed lock as fast as the steering allows (EN 16203 practice).

Case	Predicted	Measured	Verdict
Tilt table, lateral, unladen	tips at 21.9° (support triangle from the model's own CG: 4,057 kg, 0.20 m from the edge, 0.50 m above it)	uphill front tyre unloads at 20.3°, overturns at 29.3°	pass (ISO 22915-2 requires 3.4° laden; far above)
Tilt table, lateral, rated load at 3.0 m, mast back	11.5°	unloads at 9.8°, overturns at 10.6°	pass ($\geq$ 3.4°)
Tilt table, longitudinal, rated load at 3.0 m, mast vertical	5.6°	rear tyres unload at 5.1°, overturns at 6.0°	pass (ISO: 4 % = 2.3°)
Steady circle at 30° and 60° knuckle, speed rising	wheel lift near 3.6 m/s^2 lateral (the static unload)	2.5 and 1.7 m/s^2 at most, roll under 0.7°, no lift	inconclusive: the super-elastic tyres' cornering drag holds a steady circle below the lift threshold
Run-up at 17 km/h, then 30° of lock	a wheel lifts (lateral overturn hazard, NIOSH)	the inner front tyre lifts at 13.1 km/h and 5.7 m/s^2, peak roll 5.1°, no overturn	pass
Run-up at 17 km/h, then 20° of lock	a wheel lifts	lifts at 15.4 km/h and 4.5 m/s^2, peak roll 4.6°	pass

The rated load here is the H2.5A's rated moment: 2,500 kg with its centre at the rated lever, 0.970 m ahead of the front axle (a dense unit load standing at the mast end of a GMA pallet). With it the steer axle carries 746 kg against the sheet's 792 kg (from the tilt table's static tyre loads). The same 2,500 kg on a full 48 in pallet carried lengthwise, its centre 0.625 m from the fork face, is 22 % over the rated moment and leaves the steer axle 489 kg. Tipping comes out earlier than the rigid static triangle predicts (the uphill tyre unloads 7-15 % before the predicted angle; the tyres' compliance and the axle's oscillation let the centre of gravity move toward the edge first, which is the likely reason but was not measured separately). The transient run-up, not a steady circle, is what lifts a wheel, as the NIOSH fatality scenarios describe.

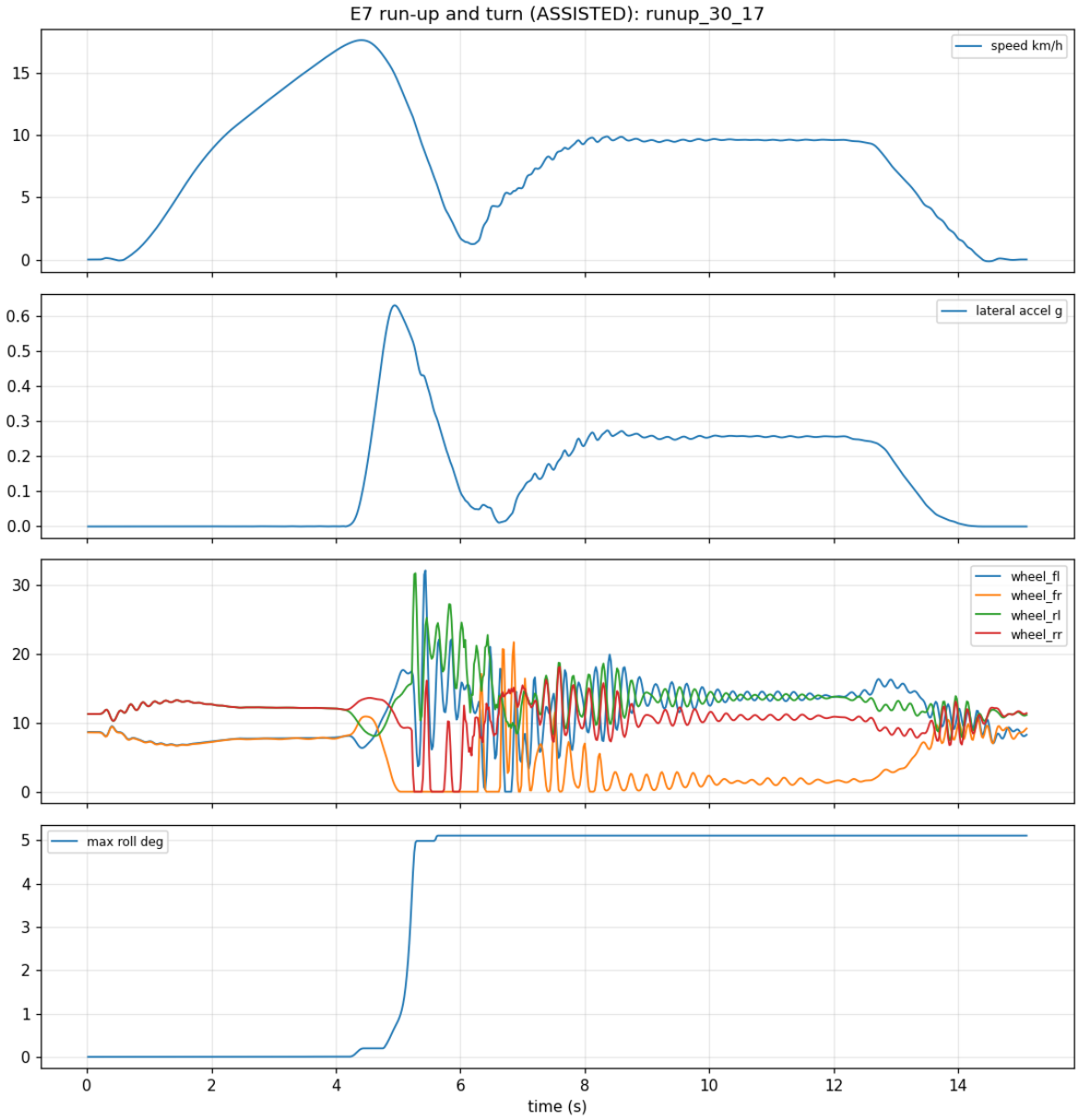


Figure 6. E7: run-up at 17 km/h, then 30° of lock: speed, lateral acceleration, the four tyre loads (the inner front tyre reaches zero) and the peak roll.

9.5 Braking (E8)

The phantom foot pushes the brake pad with a prescribed force, ramped in 0.15 s, from about 17 km/h with the forks 3° back. Laden on dry concrete the stop's plateau deceleration is 0.167, 0.252, 0.276 and 0.370 g at 300, 450, 492 and 600 N on the pad (the mean over the stop 0.152, 0.221, 0.242 and 0.309 g; stopping distances 7.55, 5.18, 4.74 and 3.72 m), with the line at 6.3, 9.6, 10.5 and 13.0 MPa and the pedal at 63, 70, 72 and 76 % of its travel: a harder push brakes harder, as predicted. The 450 N plateau, 0.252 g, meets the 0.25-0.30 g that the standard's 25 % drawbar drag gives with rolling resistance. That reproduces the brake factor's derivation, so it is reported as a fit check, not a validation (the mean, 0.221 g, falls below the band). Unladen the plateau is 0.161 and 0.338 g at 300 and 450 N (mean 0.203 and 0.303 g; 5.67 and 3.80 m), the front tyres sliding for 1.6 % and 3 % of the stop, and the stops ring at about 6 Hz. On wet concrete the laden stops match the dry ones within about 0.001 g; unladen the plateau is 0.252 and 0.371 g, the front tyres sliding for 10 % of the 450 N stop.

The laden stops pitch the truck onto its front axle. The steer wheels rise 6.7, 22.9, 55.5 and 133 mm and stay off

the floor for 0, 0.57, 1.81 and 1.78 s at the four forces, and the load slides 0.007, 1.3, 2.8 and 154 mm on the forks. So a laden stop at the standard's 450 N already lifts the steer wheels for longer than the evaluator's 0.3 s rule allows: with the rated load at the rated moment a policy must brake more gently than the standard's force. The brake sits at the standard's minimum. With 20 % more (the 1969 edition's 30 % drag) the same stop reaches 0.30 g, lifts the steer wheels 122 mm for 1.89 s and slides the load 35 mm; with the tyres' damping 1.5 times higher the lift lasts 0.53 s; across E21b's tyre-friction bracket the plateau stays 0.25 g and the lift 0.58-0.60 s.

The probe also books each stop's energy from the brake onset (brake, tyre slip, rolling, driveline) against the mechanical energy lost: the two agree within 0.7 % in every stop except the laden 600 N one (6.6 %), all inside the 10 % rule. The parking brake holds the laden truck on a 15 % grade, creeping 5.7 μm over 120 s (the parking brake is applied in the start state and no pedal is pressed).

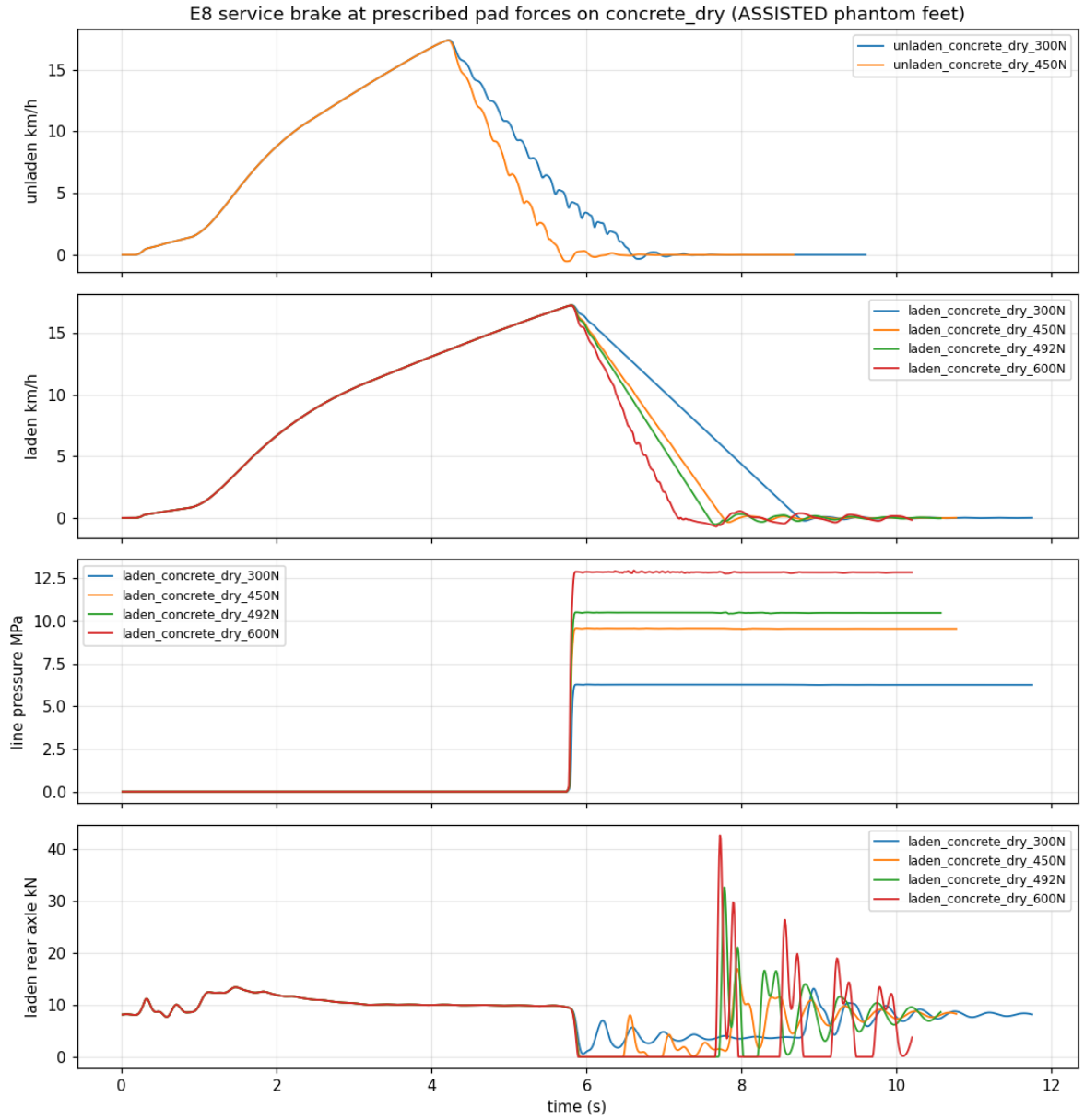


Figure 7. E8: service-brake stops on dry concrete from about 17 km/h at prescribed pad forces: speed unladen (300 and 450 N) and laden (300, 450, 492 and 600 N), the laden line pressure, and the laden rear (steer) axle's load, which falls to zero from 450 N.

9.6 Loads on the forks (E9)

A wrapped unit load on a pallet on the forks, the truck braked ever harder; the slide-onset bracket (largest peak deceleration held, smallest slid) against the $\mu = 0.35$ model's prediction: level forks held to 2.16 m/s^2, slid from 4.55 (predicted 3.43); tilted back 6° held to 2.24, slid from 5.66 (predicted 4.44); tilted forward 6° held to 2.17, slid from 4.65 (predicted 2.39). Every prediction lies inside its bracket: pass.

A loose column-stacked carton load on a pallet under rising brake force: at 0.28, 0.79 and 2.78 m/s^2 of peak deceleration no carton moved; at the full brake's 5.48 m/s^2 peak ten cartons moved more than 30 mm (51 mm at most) and none fell. The prediction, that the top cartons slide from about 2.9 m/s^2 ($\mu = 0.30$ for a sustained deceleration), fails by the probe's rule; the four brake levels place the onset between 2.78 and 5.48 m/s^2 and do not resolve it further. The probe records peaks only; the stops' sustained decelerations are lower and were not recorded, so whether the cartons' friction or the brevity of the peak holds them is not separated.

9.7 Strikes (E10)

The truck driven into a column guard at 1, 2 and 4 km/h by the autopilot. Every impact stops the truck by contact, the fork-blade tips meeting the guard first: 1.07 km/h at impact, 40 kN peak, 0.06 mm deepest penetration, rebound 0.26 km/h; 2.09 km/h, 116 kN, 0.03 mm, 0.44 km/h; 4.24 km/h, 182 kN, 0.04 mm, 0.62 km/h. The peak force grows with speed and every rebound is slower than the impact; the strike latches in every run and the evaluator's no-strike criterion fails, as it must.

9.8 Dock (E11) and ramps (E12)

Dock: the autopilot drives the laden truck along door 2's centreline and stops with its front axle over the trailer's rear sill: the sprung trailer sags 115 mm (predicted 89-127 mm, the air suspension's 3.5-5.0 in; TRAILER_K is fitted to that band, so this checks the fit); then up to 8.6 km/h and a hard stop 9 m inside. With the restraint hooked the trailer moves at most 3.0 mm away from the dock ($\leq$ 5 mm: pass). With chocks only and with neither it moves 8.9 mm, the same in both: the trailer's spring brakes hold its tyres, which never roll, so the chocks (a stop against rolling tyres) carry no force and the body moves on its suspension's fore-aft compliance. The predicted order (chocks less than nothing) is therefore refuted, and kept as such: on a trailer with its brakes set, chocks do nothing in a single stop. In every run the carried pallet's bottom deck met the trailer floor during the stop (0.24-0.25 m/s, 26-44 kN), which the evaluator counts as a strike, so the probe's no-strike check fails; the cause in the probe's driving was not isolated.

Trailer walk: five laden entries at 9.7 km/h along the door's centreline, each ended by a hard stop 9 m inside and a slow reverse onto the dock, with chocks only and then with the restraint. Chocked, the trailer moves up to 12.3 mm away from the dock during a stop and settles 3.3-3.5 mm from its spot after every one: it does not walk. Restrained, it settles 0.1 mm away (3.0 mm at most): the restraint holds. In every stop of both runs the carried pallet's deck again met the trailer floor (0.36-0.47 m/s, 28-65 kN). The prediction, that the chocked trailer's displacement grows with every entry (Hendrickson L816's dock walk), is refuted on this model: the trailing-arm air suspension (`TRAILER_ARM_SLOPE`, fore-aft axle travel per unit of spring compression, an open value) moves the body toward the nose under the laden truck, but after every stop the body returns to the same position, so nothing accumulates; which of the model's supports (the braked tandem's stick contacts, the landing gear's sand shoes) holds it there was not separated. What makes a real trailer accumulate (slip of the tyres or the sand shoes between entries, the suspension's hysteresis) is therefore not shown by this model; open (section 15).

Grades: the laden truck (rated load at travel height, facing uphill, forward selected, engine idling) released from the service brake. Neutral coast-down first, the direct test of the tyres on a grade: on 4, 8 and 13.3 % the mean acceleration down the slope is 0.189, 0.566 and 1.060 m/s^2 against g ($\sin\theta - C_rr \cos\theta$) $M/(M + \Sigma J/r_e^2)$ = 0.189, 0.566 and 1.060 from the model's own masses, wheel inertias and rolling resistance: within 0.11 % (pass). At idle in gear the truck creeps up 2 % (0.70 m), climbs 59 mm on 4 % and then nearly stops, and rolls back on 13.3 % (2.11 m, 3.6 km/h). The prediction written before running said it would roll back on 4 %; that prediction was mis-derived: the design figures (2.7 kN of stall force at idle against 1.27 kN of laden rolling resistance, which resists rolling back as much as climbing) hold the truck between 2.2 and 6.2 %, which is what the world does. A hill start on 13.3 % (accelerator to 0.6 while the brake comes off over 0.5 s) rolls back 0 mm and climbs 8.3 m: pass. The yard ramp's lip under the laden truck (from the yard demonstrations' recordings): going up the trailer's

sill sank 101 mm, the plates hung on their 3 in stop for 3.8 s and left a 24 mm step at the toe; coming down 5.5 mm (pass: under the 70-128 mm a lip fixed to the world leaves).

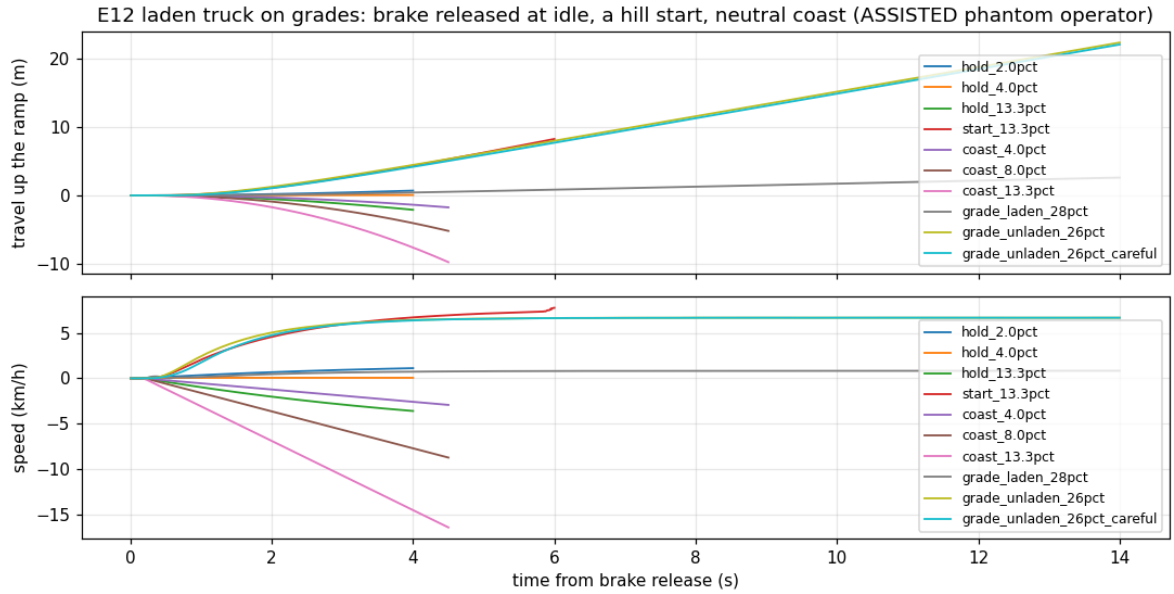


Figure 8. E12: the laden truck released on grades: at idle in gear (hold), a hill start, and coasting in neutral.

9.9 Time-step convergence (E18), the 10-minute soak (E19) and doing nothing (E20)

Convergence: halving the tick from 0.5 to 0.25 ms changes the 0-15 m time by 0.34 %, the stopping distance from 18.5 km/h by 0.21 % and the lift speed by 2×10^{-5}: pass (< 5 %).

Soak: parked 600 s, engine off, parking brake on, the rated load at 2.0 m and the H1-2 seated holding its start command. The forks creep down 34.2 mm at a steady 3.42 / 3.42 mm/min (first and second half; the lift-cylinder leakage was fitted to 35 mm, ISO 3691-1 allows 100 mm); the truck moves 0.013 mm, the load 0.0008 mm on the forks and the robot's pelvis 0.12 mm in the last 300 s; the largest speed in the last 10 s is 0.06 mm/s (carriage): pass.

Doing nothing: the robot holding its start command for 3 s and for 10 s, on 45 task-seed pairs across every task, succeeds nowhere, as predicted.

10. The seated H1-2 working the controls (E14)

SCRIPTED ROBOT: the H1-2's own motor channels (q*, dq*, kp, kd, tau through the same LowCmd decoder a policy uses), driven by inverse kinematics on the robot's MuJoCo model with clearance to the cab, reading the controls' angles as privileged state. Nothing is attached and no force acts on the truck except through contact. These runs ask one question per control: can the seated humanoid, through the action interface, physically work it? They are not policies, and where they stop short the cause is stated and attributed (the world, the robot, or the script). Each run is recorded.

How each control is worked (each technique is the result of a reach or contact study):

- Hand levers (lift, tilt, side-shift; direction stalk; parking brake): the palm laid on the handle and carried along the handle's own arc, pressing 12 mm into it; the final approach is solved free and then with clearance seeded from the free solution (a clearance-constrained path from the resting arm stalls behind the knee). The clearance pairs are every link of the arm from the shoulder out against the cab and the controls. Pairs that start at the elbow let the planner drive the left upper arm into the steering-wheel rim (129-233 N) on every leaning reach without seeing it.

- Hanging pedals (brake, inching): the ball of the foot on the pad, force-controlled: the leg holds the pad's target- travel point and adds J^T F along the pad normal, F integrated on the pedal's own travel signal; for the brake the sole is pitched 20° toe-down from the pad (flat on the pad the toe met the cowl and the ball stopped 36 mm short at 0.9 travel).
- Organ accelerator: the toe edge on the plate's upper part, the heel lifted, the sole along the toe board (the flat 26 cm foot cannot lie on the 55° plate: free inverse kinematics reaches it at every tested sole angle with no joint at a limit, but the sole meets the toe board beside the 6 cm plate).
- Steering: the spinner knob caged under the left palm (knob under the distal palm, fingers curled round it, thumb opposed), the hand carried round the column; and, as a comparison, the rim gripped hand over hand.

Control	Technique	Measured	Verdict	Cause of the shortfall
Lift lever (right palm)	palm on the knob, carried along the lever's arc	pulled to 88 % of the valve's travel: the carriage rose through its 0.96 m free lift and the main stage 0.16 m; pushed to 85 %: lowered to the stop; the lever back at −0.09° after the hand left; no neighbouring lever moved	fail by its rule (95 % of travel each way), the mast worked both ways	script (the palm's carry stops short of the lever's end)
Inching pedal (left foot)	ball of the foot, force-controlled	did not move under any target (375-400 N); after the release step the lifting foot pushed it to 0.64, and it rested at 0.56	fail	script (the ball lands 70 mm off the pad's target point)
Brake pedal (right foot)	ball of the foot, sole pitched 20° toe-down, force-controlled	0.27-0.29 for the 0.3 target (96 N); up to 0.47 under the 0.6 target, then the foot left the pad and the pedal stayed at rest for the remaining targets at 400 N	fail	script (placement 86 mm off)
Accelerator (right foot)	toe on the plate's upper part, force-controlled	0.32 / 0.42 / 0.41 / 0.41 / 0.41 for 0.3 / 0.6 / 0.9 / 0.5 / 0.2; 0.45 at most; released	fail (does not follow the targets past 0.42, nor back down)	script
Steering, spinner knob (left hand)	knob caged under the palm, hand carried round the column	one grasp: 16 / 41 / 65 / 84° of wheel for 90 / 180 / 270 / 360° of hand (the knob turns in the cage); hand-over-hand regrasps: −55° for +720° commanded (the wrong way) and −90° for −720°	one grasp turns the wheel a quarter of the hand's travel; multi-turn fails	script (lifting off the knob drags it back)
Steering, rim (left hand)	rim gripped, carried 60° per grip, 18 grips each way	−15° and +97° of wheel for ±720° commanded (both the wrong way)	fail	the hand (the Inspire grip slips on the 44 mm rim) and the script
Direction stalk (left palm)	palm on the stalk, carried along its arc	inverse kinematics with cab clearance missed the palm pose by 42-98 mm; the stalk moved 0.28°	fail	script (no reach study for the stalk)
Parking brake (left palm)	palm behind the handle, pushed along its arc at a 45° lean	lever stayed at −30.0° (applied); see below	fail	open: the robot's trunk (no waist pitch) and the asset's lever position
Key, horn	none	not attempted by this script (E14b below works them in assisted fixtures)	none	script

These are scripted motions with no assistance at all, and they fail more often than they pass: the script's foot and hand placement, not the truck, is the first limit (the assisted fixtures of E14b, below, press the same pedals and levers past the standard's forces or through their travel with the robot's own feet and fingers). In these runs the script placed the foot 28-86 mm from its target point on the pad.

The trunk while the feet work. The recorded poses of these runs, from an earlier build, show what the force records alone do not: as a foot leaves the floor for a pedal, the robot's trunk folds forward on its hips and the pelvis slides on the cushion. Read from those recordings: in the three pedal runs the trunk pitched forward to 43.5° (brake),

50.9° (inching) and 49.1° (accelerator), within 3-4 s of the foot leaving the floor, and the pelvis travelled 135-180 mm fore and aft on the cushion; in the hand-control runs the script commands a lean of 24° and the trunk overshot to 39.6° (52.5° for the parking-brake reach, which commands the deepest lean). The script commands the leg's joints only and holds no trunk posture, and its gravity feed-forward is computed with the pelvis as the fixed root, which is not the seated statics (the thighs rest on the cushion); a seated driver braces the trunk on the seat back. Cause: the script.

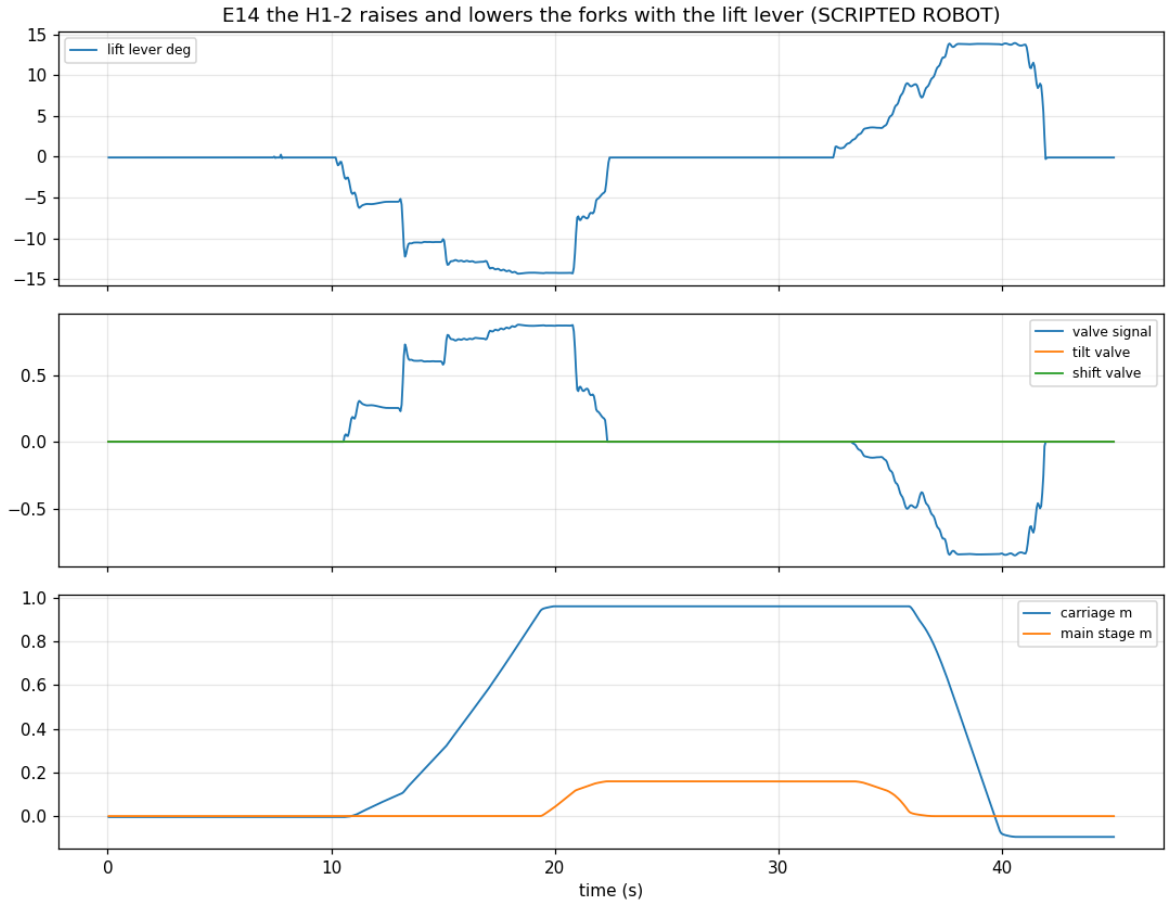


Figure 9. E14: the H1-2's right palm works the lift lever: lever angle, the valve signals (the neighbouring levers stay at zero) and the carriage and main stage.



Figure 10. From the film robot_lift (E14b, assisted fixture, recorded in an earlier build): the H1-2's right index finger has pulled the lift lever to −15° (valve +0.95, the carriage rising); the hand camera and the cab camera, the caption reading the truck's signals and the supports' forces. The scripted runs of this section are recorded but not filmed.

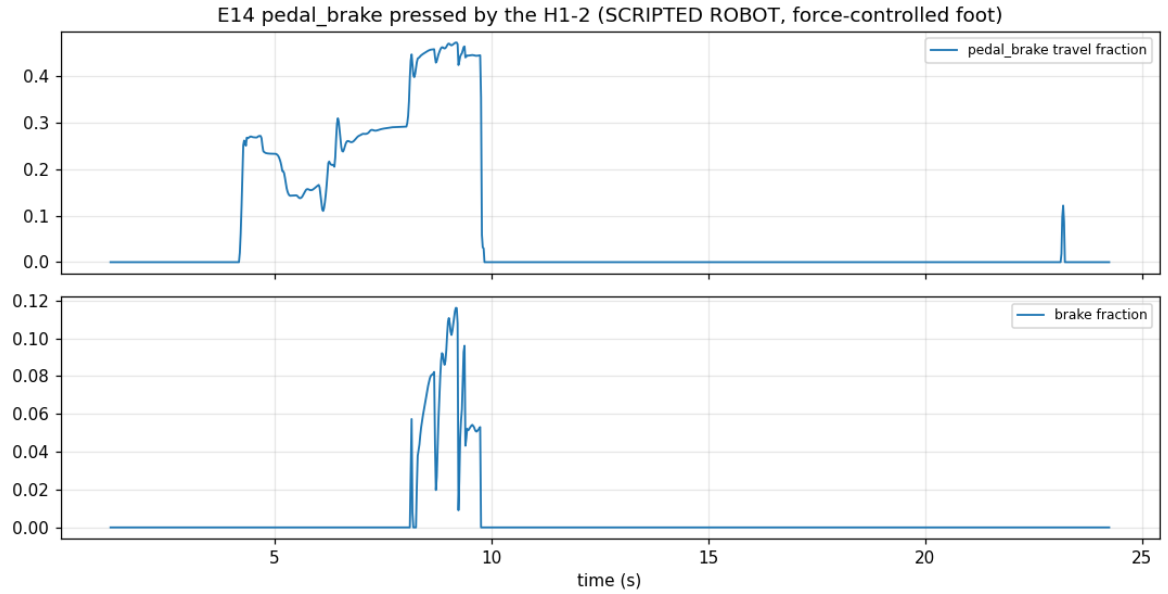


Figure 11. E14: the brake pedal's travel under the H1-2's right foot and the brake fraction the truck reads. The pedal reaches 0.27-0.29 under the 0.3 target and 0.47 under the 0.6 target; at 9.8 s the foot leaves the pad and the pedal returns to rest for the remaining targets.



Figure 12. From the film robot_brake (E14b, assisted fixture, recorded in an earlier build, before the hydraulic brake): the H1-2's right foot has pressed the service brake to 91 % of its travel (332 N of contact); the cab cameras from both sides, the caption reading the pedal's travel, the foot's contact force and the supports' forces. The pad itself is under the dash from every camera position tried.

The parking brake

Every task starts with the parking brake applied, so this reach decides whether the H1-2 can begin any task. The handle is the asset's, low at the front left of the footwell. A dedicated study separates what the robot's kinematics allow from what the scripted motion achieves:

Inverse-kinematics error to the handle (best of up to eight starts per cell):

Lean reached	Arm only, palm position	Arm only, palm facing the push, with clearance	Arm + torso, palm position	Arm + torso, palm facing the push, with clearance
28.6°	0.02 mm	10.6 mm	0.01 mm	0.10 mm
46.0°	0.03 mm	0.07 mm	0.02 mm	0.08 mm

Lean reached	Arm only, palm position	Arm only, palm facing the push, with clearance	Arm + torso, palm position	Arm + torso, palm facing the push, with clearance
51.7° (the deepest the robot holds)	0.03 mm	43.1 mm	0.02 mm	0.04 mm

Kinematically the handle is within reach at every lean once the torso joins the reach. The best physical attempt did not get there: the palm came no closer than about 58 mm to the handle and the lever did not move from −30° (its over-centre point is at −18°). Leaning in, the robot's torso bore on the spinner knob at up to 258 N; carried along the release arc, its right shoulder bore on the tilt lever at 290-337 N and its torso on the knob at 31-57 N.

The release is not demonstrated. What blocks it is measured: the H1-2 has no waist pitch, so reaching the handle needs the whole trunk pitched on the hips, which brings the trunk and the shoulders against the spinner knob and the mast levers. Section 15 states what this means for the tasks and what remains to check against the real truck.

Assisted robot fixtures: can the cab be worked at all (E14b)

ASSISTED ROBOT FIXTURE: one control per fixture, worked by the H1-2's own fingers or feet. The pelvis is held by an external spring-damper at a pose prescribed in the truck's frame (the seated pose, or a lean for a reach), and the working wrist or ankle is pulled toward its planned path by a bounded force (hand ≤ 60 N and 6 N·m, foot ≤ 150 N and 20 N·m); both react on the chassis, so they are forces from outside the robot, labelled as such. The planner is privileged (inverse kinematics on the robot's model with the controls' poses); the motors are commanded through the policy's motor interface. No control's joint is written, nothing joins robot and control, no phantom force acts on a control: each control moves only through the contact forces of the robot's links, which every record measures, together with every other contact between the robot and the cab ("bystander": any link pressing the cab or another control with more than 20 N). These runs answer one question per control, whether the cab lets the seated H1-2 work it by contact at all, which the scripted motions above leave open. Seven fixtures were run on the current build; the key, parking brake, direction stalk and horn rows are from an earlier build (before the hydraulic brake, the guard at the sheet's height and the 5° start lean), marked below.

Control	Result	Criteria (passed / all)	Peak contact on the control (N)	Peak pelvis support / limb guide (N)	Pelvis vs seat, max / end (mm)	How the fixture works it
ignition key (earlier build)	**fail** (on, start, engine_started, back_on, bystander)	2 / 7	0	490 / 60	101 / 1	the right index fingertip pressed on the key's head, turning it by friction
parking brake (earlier build)	**fail** (applied_again, bystander)	4 / 6	135	757 / 60	127 / 1	the left index finger (pad under the handle end to release, fingertip pad on its top-left edge to apply)
direction stalk (earlier build)	pass	6 / 6	15	337 / 60	72 / 2	the left index finger pad
steering (spinner knob)	**fail** (right_lock, straight, own_body, bystander)	2 / 6	1,802	238 / 60	72 / 1 (prescribed 25° lean)	the left hand caging the spinner knob, orbited round the column
lift lever	pass	7 / 7	26	33 / 11	0.1 / 0.0	the right index finger (pad on the handle's top for the pull, fingertip on its end face for the push)
tilt lever	pass	7 / 7	111	112 / 38	2.4 / 0.4	the right index finger, as the lift lever
side-shift lever	**fail** (pulled: 89 % against 90 %)	6 / 7	34	87 / 35	2.0 / 0.5	the right index finger, as the lift lever
accelerator	pass	8 / 8	90	913 / 150	8.2 / 6.1	the right foot's fore
service brake	pass (782 N on the pad, 78 % of the travel)	7 / 7	782	458 / 150	5.5 / 5.5	the right foot's toe
inching pedal	**fail** (full, released, bystander)	4 / 7	989	1,475 / 150	10.2 / 6.7	the left foot's fore

Control	Result	Criteria (passed / all)	Peak contact on the control (N)	Peak pelvis support / limb guide (N)	Pelvis vs seat, max / end (mm)	How the fixture works it
horn (earlier build)	pass	6 / 6	8	25 / 12	0 / 0	the left index and middle fingertips

On the current build the lift and tilt levers, the accelerator and the service brake pass every criterion. The lift lever rises 0.49 m and lowers 0.70 m under the finger, and the tilt lever is pulled to 92 % of its travel. The side-shift lever is pushed through its travel but pulled to 89 %, 1 % short of its 90 % rule, with the mast answering both ways. The accelerator goes to full travel, is held at half for 2 s and is released. The service brake passes with 782 N on the pad, past the standard's 450 N, at 78 % of the travel: the hydraulic brake's floor stop needs 1,800 N, so the criterion asks for the standard's force or 95 % of the travel. The inching pedal goes to 0.50 at most, and after the foot leaves its pad the pedal still stands at 0.22; the left ankle bore on the brake pad at 851 N and the left hip on the cab at 1.94 kN. The steering fixture caged the knob and orbited the hand round the column: it turned the wheel 1,335° (3.7 turns) to the left lock (knuckle −79.1°) while the hand orbited 3,191°, but turning back it came 1,062° only, the knuckle ending at −17°; the left shoulder pressed the torso at 385 N and the fingers pressed the horn button at up to 326 N.

In the earlier build the direction stalk went through F, N, R, N and the horn sounded; the parking brake was released (the left index finger's pad under the handle end, 135 N, 14.9 N·m about the hinge) but not applied again: the best plan to the handle's end was 21 mm off and the finger never touched it, and the handle's apply side lies inside the asset's cowl when released, so a hand reaches only its underside, which pushes toward released. The key failed: the right index fingertip never touched its head, with the wrist guide at its 60 N cap and the hand up to 186 mm and 36° off its plan. With them, six of the eleven controls pass every criterion. Open: the key, the inching pedal and the steering's return on the fixture's planning (cause: the fixture), the side-shift pull by 1 %, and the parking brake's re-apply on the lever's released pose against the cowl, which needs checking against a real H2.5A (cause: open, world or asset).

11. Every task through the unmodified evaluator (assisted demonstrations, E15)

ASSISTED: an autopilot reading privileged state (pose, knuckle angle, fork height, the loads' poses) works phantom hands and feet at the truck's own controls, with the H1-2 seated and holding its start command; the evaluator is the one a policy faces, unmodified. These runs show that each task can be completed through the truck's controls in this world at its real difficulty; they are not policies, and the robot's own motors drive nothing here (section 10 covers what the robot itself can operate, section 4.4 what the kit drove). Seed 0 of every task, and one more seed of each task whose target varies, from the test split (a target different from seed 0's: another staging lane, floor bay, door and side, bale door, start, direction, drop zone or route; seed 902 of load-trailer is an unsecured trailer, whose job is to wait at the door short of the leveler). Seed-0 runs record a film.

Task	Seed	Difficulty	Success	Score	Sim time / limit (s)	Wall / sim
aisle-cruise	0	easy	yes	1.00	27.4 / 120	9.9
aisle-cruise	900	easy	yes	1.00	29.8 / 120	10.0
carry-loose-stack	0	very hard	yes	1.00	74.7 / 240	21.2
carry-loose-stack	900	very hard	yes	1.00	77.1 / 240	23.0
carry-to-floor-bay	0	medium	yes	1.00	64.5 / 180	9.5
carry-to-floor-bay	900	medium	yes	1.00	71.5 / 180	9.3
drive-to-pallet	0	medium	yes	1.00	20.3 / 120	11.1
drive-to-pallet	900	medium	yes	1.00	21.2 / 120	11.1
feed-conveyor	0	hard	yes	1.00	100.2 / 180	8.3
feed-conveyor	901	hard	yes	1.00	98.0 / 180	9.7
load-bales	0	very hard	see below	0.909 (earlier scoring)	337.0 / 480	11.4

Task	Seed	Difficulty	Success	Score	Sim time / limit (s)	Wall / sim
load-bales	900	very hard	see below	0.909 (earlier scoring)	317.4 / 480	12.3
load-trailer	0	hard	yes	1.00	113.5 / 240	12.7
load-trailer	902	hard	yes (unsecured: waited at the door)	1.00	16.1 / 240	14.9
load-trailer-chocked	0	hard	yes	1.00	113.3 / 240	12.2
load-trailer-chocked	900	hard	yes	1.00	113.5 / 240	13.4
put-away-rack	0	hard	yes	1.00	92.7 / 240	10.0
ramp-down-loaded	0	hard	yes	1.00	62.1 / 240	11.5
ramp-down-loaded	904	hard	yes	1.00	66.0 / 240	11.7
ramp-up-loaded	0	hard	yes	1.00	103.8 / 240	11.6
ramp-up-loaded	900	hard	yes	1.00	105.4 / 240	12.8
start-and-park	0	easy	yes	1.00	27.1 / 90	10.7
start-and-park	900	easy	yes	1.00	34.3 / 90	10.4
unload-trailer	0	very hard	yes	1.00	128.6 / 300	11.8
unload-trailer	900	very hard	yes	1.00	128.2 / 300	13.0
yard-route	0	easy	yes	1.00	33.3 / 150	11.4
yard-route	901	easy	yes	1.00	33.0 / 150	10.4

25 of the 27 runs succeed at 1.00. The two load-bales records were scored by an earlier version of the evaluator, which gave the fraction of criteria met (0.909) and failed only two_bales_at_nose: one bale sat on its strips with its front face at the nose wall (a gap of about 0) and was read as outside the trailer. The current evaluator allows a resting face 5 mm into the wall's plane and scores load-bales by its progress reward (section 7). Replays on the current code succeed at 336.25 s (seed 0) and 317.65 s (seed 900). The load-bales plan backs out straight before turning, so it does not sweep its load through the staged row. The autopilot drives the yard lane at 1.5 m/s, because at 2.2 m/s its steering loop weaves on these tyres; the operator's loop is tuned, not the world. Wall / sim is the run's wall time over simulated time on a shared host, including the film recording.

Put-away seeds: eighteen more put-away-rack seeds across the splits (1, 3, 4, 7, 10, 13, 800, 801, 806, 807, 809, 900, 901, 902, 903, 907, 908, 950), each a different rack location, all succeed at 1.00 in 67.7-94.1 s. Their nineteen start states (with seed 2) all pass E0's rest check: drift at most 0.066 mm, the largest contact at rest 2.29 kN.

Predeclared negative controls

Each negative is the same plan with one operator error, declared before running with the criterion it must fail. A negative that does not fail its criterion is a finding about the world or the evaluator, not a pass.

negative (task, seed 0)	the one operator error	must fail: measured	as predicted	also failed
no-parking-brake (start-and-park)	parked without setting the parking brake	parking_brake_set: not applied	yes	none
forks-high (drive-to-pallet)	blades 50 mm above the GMA opening band	blade_height: 85 mm above the floor (20-60 mm allowed)	yes	none
fast-turn (carry-to-floor-bay)	15 km/h set point, no slowing for the turns	speed_limit: 12.3 km/h (8.8 allowed)	yes	a wheel off the floor 0.92 s; the load touched the floor 11 times; parked outside the bay lines, 6.8° off square
ramp-too-fast (ramp-down-loaded)	down the 13.3 % ramp at 9 km/h	slow_on_ramp: 9.5 km/h on the ramp (6 allowed)	yes	speed_limit
stack-hard-stop (carry-loose-stack)	a full service-brake stop from 5 km/h with the loose stack	cartons_in_place: 11 mm (30 allowed): **passed**	**no**	none: the run succeeds
chocked-hard-stop (load-trailer-chocked)	into the chocked trailer at 5.8 km/h, a full service-brake stop near the nose, 2.5 m short of the nose pallets	trailer_movement: 5.7 mm (25 allowed): **passed**	**no**	none: the run succeeds

negative (task, seed 0)	the one operator error	must fail: measured	as predicted	also failed
chocked-nose-collision (load-trailer-chocked)	into the chocked trailer at 5.8 km/h, braking too late, so the load strikes the nose pallets	no_strike: 5 strikes	yes	the load on the floor (20 contacts), not in the trailer behind the nose pallets, not at the side wall; the trailer moved 14.9 mm (25 allowed)

Five of seven failed exactly as predicted. Two predictions are refuted, and kept as such:

- **stack-hard-stop.** With the brake pedal fully down from 5 km/h the top cartons move 11 mm, well short of the 30 mm that counts as out of place. E9's stack under rising brake force (section 9.6) shows the onset of sliding lies above a 2.78 m/s^2 peak. The stop is as hard as this truck's brakes and tyres make it from that speed; a spill needs a harder event (a strike, a fast turn).
- **chocked-hard-stop.** The trailer held only by chocks moved 5.7 mm under the stop, against the predicted more than 25 mm. E11 (section 9.8) shows why it cannot move more in one stop: the trailer's spring brakes hold its tyres, which never roll, so the chocks carry no force, and the body moves only on its suspension's fore-aft compliance (8.9 mm under a hard stop from 8.6 km/h). Even the collision of the seventh negative moved it only 14.9 mm.

12. Stability at rest and under legal abuse

12.1 Every start state at rest (E0)

Each task's start state is built, settled by physics (construction only, `World.settle`: every body still within 0.5 mm and 0.5 mrad over 0.5 s), checked (`env.start_check`: settled, operator seated, every wheel loaded, no penetration over 10 mm, no contact over 100 kN), then held 120 s with nothing acting except the robot's start command and the truck's own mechanisms. Rejected if any body drifts 2 mm over the last 60 s or travels 1 mm in the last 10 s (the carriage and carried loads judged relative to the forks, because the lift cylinders leak by design), if a signal changes, or if the forks drop faster than ISO 3691-1's 100 mm in 10 min.

Task	Worst drift, last 60 s (mm)	Worst drift, whole hold (mm)	Largest travel, last 10 s (mm)	Fork drop (mm / 10 min)	Largest contact force at tick 0 (N)	Signal changes
aisle-cruise	0.018 (pelvis_in_seat)	0.010 (pelvis_in_seat)	0.009	2.4	332	0
carry-loose-stack	0.039 (load:carried_c21)	0.354 (load:carried_c26)	0.011	6.1	679	0
carry-to-floor-bay	0.024 (pelvis_in_seat)	0.017 (pelvis_in_seat)	0.016	7.7	996	0
drive-to-pallet	0.055 (chassis)	0.114 (chassis)	0.013	0.2	372	0
feed-conveyor	0.031 (pelvis_in_seat)	0.020 (pelvis_in_seat)	0.013	16.7	2853	0
load-bales	0.328 (load:bale_0)	0.372 (load:bale_0)	0.118	2.4	1530	0
load-trailer-chocked	0.138 (pelvis_in_seat)	0.207 (pelvis_in_seat)	0.052	8.7	1814	0
load-trailer	0.111 (pelvis_in_seat)	0.181 (pelvis_in_seat)	0.033	7.9	1811	0
put-away-rack	0.041 (pelvis_in_seat)	0.035 (pelvis_in_seat)	0.010	8.6	2243	0
ramp-down-loaded	0.131 (pelvis_in_seat)	0.174 (pelvis_in_seat)	0.054	9.0	1267	0
ramp-up-loaded	0.108 (pelvis_in_seat)	0.193 (pelvis_in_seat)	0.037	10.9	1699	0
start-and-park	0.055 (chassis)	3.463 (pelvis_in_seat)	0.045	0.2	322	0

Task	Worst drift, last 60 s (mm)	Worst drift, whole hold (mm)	Largest travel, last 10 s (mm)	Fork drop (mm / 10 min)	Largest contact force at tick 0 (N)	Signal changes
unload-trailer	0.110 (pelvis_in_seat)	0.139 (pelvis_in_seat)	0.043	2.4	1839	0
yard-route	0.129 (pelvis_in_seat)	0.208 (pelvis_in_seat)	0.037	2.4	350	0

All fourteen pass (three of them also recorded, in an earlier build, for the time-lapse films of section 14). "Worst drift" is the largest displacement of any body over the window (the carriage and carried loads relative to the forks, whose cylinders leak by design), so a body that moves out and back can show more over the last 60 s than over the whole hold. The largest motion in the last minute is a bale in load-bales at 0.33 mm. The seated robot's pelvis moves at most 0.14 mm in the last minute; in start-and-park it settles 3.5 mm on the cushion early in the hold and then stays. The nose pallets of the chocked trailer stay put (0.005 mm in the last minute) because the convex collision tolerance is 10^{-8} m: at 10^{-6} m the convex narrow phase drops resting pallets for single ticks and they creep 1.2-2.0 mm a minute on the trailer floor. The fork drops are the lift-cylinder leakage at each task's load and height (section 3.6). E0 holds the H1-2; the kit's start state was checked only as part of E14k (it settles and its ballast closes the seat switch).

12.2 The seated robot (E13)

At rest for 120 s the H1-2's pelvis moves 0.013 mm in the last 30 s relative to the seat (0.022 mm over the whole hold), the seat switch stays closed, its largest joint torque is 12.5 % of its limit and the hip point moves 0.004 mm.

Two stops with the robot holding its start command and the lap belt fastened, each made with the service brake held at 1.19-1.22 times the torque of the standard's 450 N pedal force from 0.5 s to 4 s after the press (the rule: seat switch closed throughout, the hip point, the body's position on the cushion, within 50 mm of the seat, the robot never meeting the overhead guard above 130 N, and the brake at least at the standard's torque). The hip point is the midpoint of the hip-pitch axes: the H1-2's pelvis frame sits 0.163 m above them, so the pelvis frame's offset mixes the body's position on the cushion with the trunk's pitch, and is reported, not judged.

run	stop	hip point, max / end	pelvis frame, max	trunk pitch, max	belt	head to guard
ride: figure-8 up to 11.8 km/h, then a stop from 3.2 m/s	0.66 g peak, 0.32 g sustained	20 / 12 mm	39 mm	11.5°	locked 1.62 s (pendulum 41° against its 24.2° trip), 673 N	27.6 mm at the closest, 0 N
a straight stop from 3.3 m/s	0.66 g peak, 0.33 g sustained	18 / 7 mm	36 mm	10.6°	locked 0.96 s, 522 N	27.3 mm, 0 N

Both runs pass. Both are hard stops, with the brake held above the standard's torque. The robot stays on the seat with its hip point within 20 mm; the forward motion seen in the films is mostly the passive trunk pitching on its hips (11.5° at most), which a seated human would brace against with the feet, the hands and the back. The belt's retractor locks in both stops, its pendulum tipped past the 24.2° trip (0.45 g, UN R16), and the belt then holds the pelvis at 522-673 N. The head comes within 27 mm of the overhead guard and never touches it.

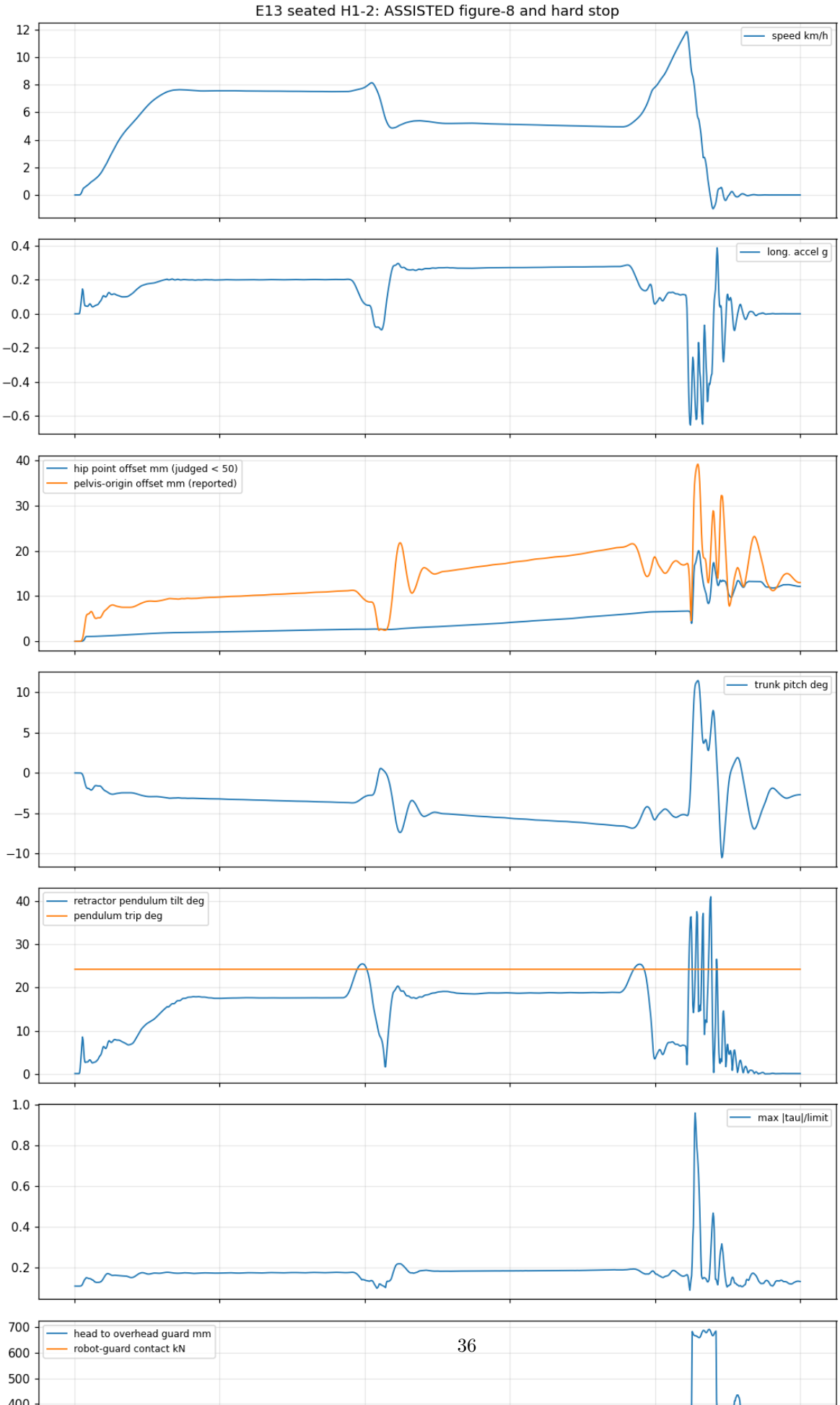

E13 seated H1-2: ASSISTED figure-8 and hard stop
speed km/h
long. accel g
hip point offset mm (judged < 50)
pelvis-origin offset mm (reported)
trunk pitch deg
retractor pendulum tilt deg
pendulum trip deg
max |tau|/limit
head to overhead guard mm
robot-guard contact kN

Figure 13. E13: the H1-2 through an assisted figure-8 and a hard stop: speed and the seat's fore-aft acceleration, the hip point and the pelvis frame relative to the seat, the trunk's pitch, the belt retractor's pendulum against its trip angle, the largest joint torque against its limit, the gap from the head to the overhead guard with any robot-guard contact, and the belt's tension (its 7 N retraction, then the locked belt's load).

12.3 Legal abuse (E17)

Random but legal Unitree LowCmd commands through the backend's own action path (each command decoded as a policy's JSON is, then validated; position targets at the joint limits, full gains, feed-forward torques of either sign, joint-speed targets at the motor's limit, on all motors or one limb, held 2-100 ms), with the truck parked and idling in gear. Rejected if the world faults, a body exceeds the unphysical-speed guard, or a state goes non-finite. Eight seeds of 60 s each (60.0-60.1 s simulated):

seed	commands (refused)	faults	fastest robot link	fastest spin (rad/s)	seat switch open	truck controls the thrashing robot moved	wall / sim
0	2,012 (0)	none	17.9 m/s	249	13.6 s	parking brake, accelerator 0.50, brake 0.31, inching 0.29, direction, all three levers	6.0
1	1,803 (0)	none	19.4 m/s	129	24.7 s	parking brake, accelerator 1.0, direction, all three levers	6.0
2	1,862 (0)	none	17.8 m/s	172	9.9 s	parking brake, accelerator 0.99, brake 0.67, inching 0.96, direction, horn, levers	6.1
3	2,004 (0)	none	20.2 m/s	94	11.7 s	parking brake, accelerator 0.89, brake 0.76, inching 0.88, levers	6.0
4	1,950 (0)	none	18.2 m/s	123	15.7 s	parking brake, accelerator 0.76, brake 0.85, inching 1.0, direction, horn, levers	6.3
5	1,837 (0)	none	18.9 m/s	199	12.4 s	parking brake, accelerator 0.99, brake 0.74, inching 0.37, direction, horn, levers	6.3
6	1,930 (0)	none	17.1 m/s	169	35.3 s	parking brake, brake 0.27, direction, levers	6.3
7	1,899 (0)	none	25.0 m/s	167	15.5 s	parking brake, accelerator 0.34, brake 0.44, inching 0.62, direction, horn, levers	6.7

Passed: no fault in 8 × 60 s of violent legal commands, the evaluator answering after each. The fastest robot link reached 25.0 m/s, under the 54 m/s guard, so the guard's value is not what passes this test. The constraint solver's largest final gradient per seed is 3.8-8.6 × 10^{-4}, and no tick of any seed ended above 10^{-3}. The thrashing robot leaves the seat (switch open 10-35 s of the 60) and works every control within its reach, a measure of how crowded the seated robot's reach is.

Three settings of the world's numerics and the hand's model carry this pass. (1) The constraint solver's line search runs MuJoCo's default 50 iterations. Capped at 20, it runs out on hard ticks: the solve stalls with its gradient at 60 and returns unbalanced forces (a shoulder limit needing 61 N·m got 13 kN·m). A tick whose solve ends unconverged faults the world instead of continuing. (2) The Inspire hand's fingers are micro servos driving lead screws through linkages, whose rotor inertia times the square of the reduction spans 0.003-0.03 kg·m^2 per finger (neither number is published, so the entry is open); the ledger takes the range's geometric middle, 0.01 per finger, split between the driver joint and its linkage follower (`ROBOT_ARMATURE`). (3) The equalities that tie each finger's linkage follower to its driver have a steel linkage's 4 ms (8 ticks); the engine's default 20 ms spring lets a struck follower swing against its driver. In an earlier build the finger inertia's part was confirmed by running the same test with 1 × 10^{-4} kg·m^2 per joint, a value with no basis: seed 1 faulted at 2.7 s, the right little finger's proximal joint at 144 rad/s, 288 times its motor's no-load speed (the joint-speed fault). The pass did not hold across the whole open range there either: at its low end (0.003 kg·m^2 per finger, 1.5 × 10^{-3} per joint) seed 0 ran its 60 s without a fault, but seed 1 faulted at 6.1 s, the left thumb's proximal yaw joint at 105 rad/s, over the 100 rad/s floor of the joint-speed fault. These range runs were not repeated on the current build. Whether that is a world defect or a real thumb's behaviour is open: a lead-screw finger drive is expected to resist being back-driven (an expectation from the drive type, not a measurement), which this model does not represent (its fingers are back-driven through their rotor inertia alone), so the low end of the range may well not be where the real hand sits; a back-driving model of the finger drive is the cause-level next step. Cause: open (the finger drive's model).

12.4 A 10-minute parked soak (E19)

The laden truck parked for 600 s with the rated load raised to 2.0 m and the H1-2 seated holding its start command. Predicted: a fork drop of 20-50 mm in 10 minutes at a steady rate, the truck under 1 mm, the load on the forks under 1 mm, the pelvis under 2 mm over the last 300 s.

quantity	measured
fork drop in 600 s	34.2 mm
drop rate, first / second half	3.42 / 3.42 mm/min
chassis drift	0.013 mm
load relative to the forks	0.0008 mm
pelvis drift, last 300 s / whole soak	0.119 / 0.200 mm
fastest body in the last 10 s	carriage, 0.06 mm/s

Passed. The drop is the lift cylinders' leakage, whose coefficient was set from the stated 35 mm in 10 minutes (section 3.6), so the 34.2 mm checks the integration (a steady rate, no other path), not an independent value; ISO 3691-1 allows 100 mm.

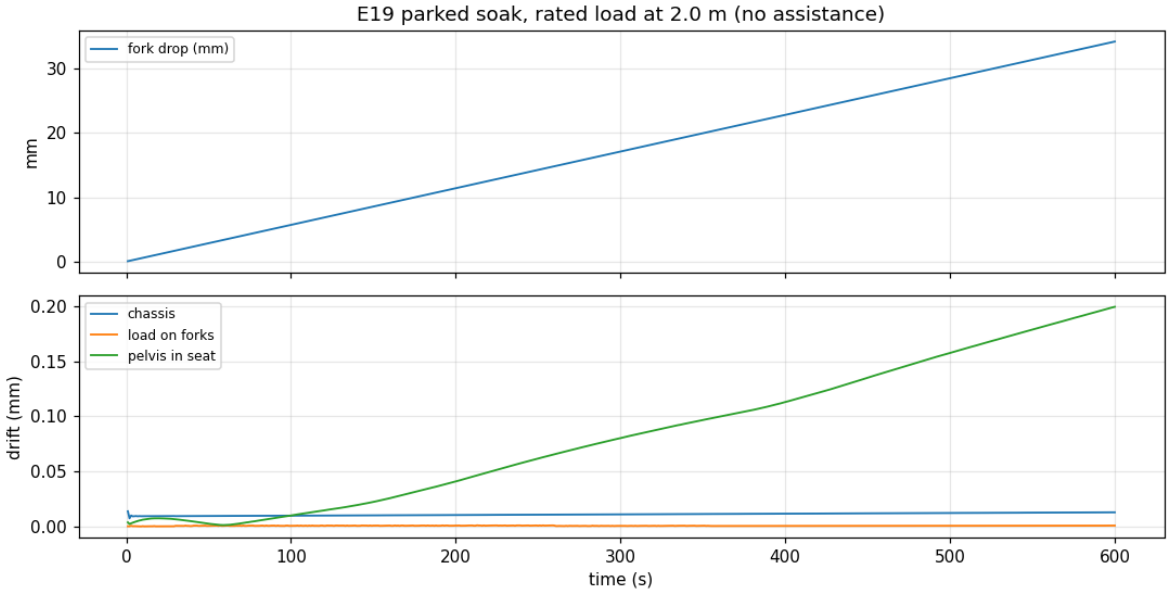


Figure 14. E19: fork drop and drifts over the 600 s parked soak.

12.5 Warnings inspected

- `mujoco_warp ... MULTICCD is enabled, but the scene contains CCD pairs without multicontact support: (CAPSULE, CYLINDER), (CAPSULE, MESH), (CYLINDER, CYLINDER), (CYLINDER, BOX), (CYLINDER, MESH). At most 1 contact will be generated for these pairs.` Printed once per world build. The capsules are the 16 segments of the steering-wheel rim; the cylinders are the four tyres (which never touch drivable surfaces through MuJoCo; they meet pallets, curbs and walls), the conveyor rollers, the coil, the horn button and two H1-2 shapes. One contact per pair matters where a single pair must resist a moment; here none does: a pallet on the conveyor rests on several rollers at once, a hand on the rim touches several segments, and a tyre against a wall or curb is a point contact in reality too.
- No other warning is printed by the experiment runs; MuJoCo's own warnings (contact or constraint buffer overflow, bad inertia, unstable simulation) are turned into world faults (`world._check`) and none is recorded in the 143 current-build records behind this report.

13. Runtime (E16)

Measured for each offered robot on a shared host (16-CPU quota, one RTX 5090 shared with other sessions; every row carries the 1-minute load and the GPU's utilisation it ran under; no quiet window opened, so every figure is a shared-host figure and a quiet host would be faster). These are the numbers the backend declares in its capabilities (`RUNTIME`, `TRACE`).

Through the served interface, as a policy's client sees it: `gar run forklift` with the policy key and the head or cab camera, every request timed with a wall clock around the HTTP call, 100 requests per command length on each of start-and-park, carry-loose-stack and put-away-rack (300 per row). The ratio is wall seconds per simulated second.

Robot (1-minute load)	Command	Median	p99	Worst	Requests over 200	Sustained (Σ wall / Σ sim)
H1-2 (12-53)	10 ms	31.1	180.8	276.1	1 of 300	62.3
H1-2	20 ms	65.7	109.5	144.7	0 of 300	55.3
H1-2	0.1 s	25.3	44.3	49.8	0 of 300	26.7
Kit (15.5-50.6)	10 ms	36.4	193.3	377.4	3 of 300	69.9
Kit	20 ms	76.8	151.7	209.5	1 of 300	63.0
Kit	0.1 s	24.1	43.5	57.4	0 of 300	26.8

The backend declares the shortest command none of whose requests exceeded 200 and from which every longer command stayed inside 200 at its 99th percentile (`shortest_command_within_200x_s`): 20 ms for the H1-2 and 0.1 s for the kit. It does not bound every request: on this saturated shared GPU the slowest single requests took 2.8-3.8 s for a 10 ms command, 2.9-4.2 s for a 20 ms one and 5.0-5.7 s for a 0.1 s one, and nothing in the environment bounds such stalls. Short commands cost more because a command that crosses a 30 Hz frame boundary renders a new camera frame inside its request: a 10 ms command crosses one about every third command, a 20 ms command two in three and a 0.1 s command always, and a new frame costs a median 1.09 s (H1-2) and 1.23 s (kit), p90 1.33 / 1.52 s, up to 2.48 / 2.35 s. That is why 20 ms commands cost more per simulated second than 10 ms ones. Every request also pays the action history's synced commits (median 0.158 / 0.127 s). GPU utilisation ran at 87-100 % throughout.

Physics alone through the served backend (owner key, 1 s commands, median wall s per simulated s): for the H1-2, 6.3 (start-and-park) to 10.1 (ramp-down-loaded) on thirteen tasks and 19.9 on carry-loose-stack (30 loose cartons); for the kit, 5.0-8.2 on start-and-park, put-away-rack, load-trailer and ramp-up-loaded and 18.7 on carry-loose-stack. In process with the trace on (0.5 s commands, all 14 tasks): 5.9-20.2 for the H1-2 and 4.9-18.8 for the kit, carry-loose-stack the slowest. At 6.3 wall s per simulated s a 0.5 ms tick costs about 3.2 ms: the CPU MuJoCo step with the truck's systems (tyres, drivetrain, steering, hydraulics) evaluated in Python every tick.

Render worker and memory. The render worker stays warm 600 s after each request (`renderer_options idle_release_s`). A request that has to start a new one (a session's first after `gar run`, or the first after a longer pause) took 71-99 s on this host at load 14-23: five requests of a cold program took 73-99 s, and a request after a 660 s pause 71 s. Runs at higher load measured up to 162 s. The new worker starts beside the command's physics, so only a command whose physics lasts as long hides the start: a 5 s command still waited 62-65 s. This was measured with the H1-2; no cold-start measurement exists for the kit. `gar run` with the camera took 91-162 s (H1-2) and 67-127 s (kit). A session with the camera holds a render worker of 6,040-8,170 MiB (H1-2) or 6,338-8,040 MiB (kit) of GPU memory, and its server and worker 6,136-6,797 MiB (H1-2) or 5,541-5,846 MiB (kit) of host memory (PSS; resident 6,776-7,324 and 6,186-6,506 MiB).

The per-tick trace (owner diagnostics, on by default), measured in process, parked on every task and driven by the assisted autopilot:

Robot	Trace time per 0.5 ms tick	Share of a traced command's time	Payload per simulated s (on disk)	Simulated time the 256 MiB budget keeps
H1-2	0.19-0.41 ms	4.1-10.4 %	3.37-6.17 MB (4.27-8.41 MB)	44-80 s
Kit	0.17-0.59 ms	3.5-18.5 %	2.81-5.54 MB (4.27-8.4 MB)	48-96 s

The share is larger when physics runs faster. One maximum command, 30 s on carry-loose-stack with the trace on, took 594 s of wall time and kept all 60,000 samples, none evicted (181 MB of payload, 68 % of the budget; 256 MB

on disk). Older samples are evicted and counted. The client timeout the backend declares, 1,800 s, covers such a command plus a cold render worker and the request's fixed cost (about 700 s here), with room for the higher loads measured (physics up to 38-47 and worker starts up to 162 s give 1,300-1,600 s).

The served record behind these figures holds 46 `gar run` sessions and 3,153 timed requests, across both robots and both keys; every request was answered 200.

14. Rendering and visual review

Images are rendered by Isaac Sim 5.1 RTX path tracing in the render-only worker from the same state the physics holds (the worker never advances simulation time). Every camera meters its own frames, as the H1-2's D435i does (exposure from 0.1 to 33 ms and an analogue gain up to 16×, about 6 stops either way of the preset's base exposure); the policy's head camera is rendered live at each observation. Every still and film below was rendered from recordings of an earlier build of this world and looked at. Details such as the overhead guard's height, the robot's start posture and the numbers in the films' captions are therefore the earlier build's, not those of sections 9-12. No film of the driving-robot kit exists. What was seen, including faults:

Scene stills (rendered at 32 samples)

Each is a task's start state built and settled as a reset builds it.

- `warehouse_overview.png`, `warehouse_side.png`: three rows of selective racking (A-C, blue uprights, orange beams, aisle and row signs) holding carton loads on pallets, the painted aisle, floor-bay and stop lines, the production edge at the back wall (the roller conveyor feeding a green press), wall fixtures (fire point, signs, bins); the truck carrying a wrapped load on a pallet with the H1-2 seated.
- `dock_*.png`: four numbered doors with levelers, door 1 raised on its trailer's lit interior, staging lanes painted on the floor, empty pallets stacked along the wall, door controls and signal lights beside each door.
- `yard_*.png`: asphalt with the painted lane and drop zone, three parked trailers, the yellow mobile ramp with its grating deck and curbs rising into a trailer on its landing gear, cones, a perimeter fence and trees under the HDR sky.
- The cameras (`warehouse_head`, `_forks`, `_cab`, `_hands`, `_chase`, `_pedals`): the policy's view over the wheel, the spinner knob and the lever console with the mast and its chains ahead; the blades in the pallet's openings under a stretch-wrapped load with its shipping label; the H1-2 on the cushion with its hands on its thighs, the LPG cylinder behind; from above the counterweight through the overhead guard's bars down an aisle; the shins in the footwell beside the pedals and the data plate. The dash overhangs the pedal pads from every camera position tried, so a pedal film shows the leg working and its caption reads the pedal's travel.
- Details (`ramp-up-loaded_lip.png`, `_stands.png`, `start-and-park_cab_key.png`, `_cab_park.png`): the ramp's four lip plates lying on the trailer floor at the sill; the two stabilizing jacks under the trailer's nose beside its landing gear; the key standing proud of the dash face; the parking-brake lever beside the steering column, its handle top clear above the cowl.



Figure 14b. Details rendered from the start states (an earlier build): the yard ramp's hinged lip plates resting on the trailer floor, the trailer's nose stands, the key in the dash, the parking-brake lever.

Films

Rendered from the runs' recordings (every body's pose at 25 Hz) at 25 frames per second, two camera tiles side by side, each film's speed chosen so that it lasts about ten seconds (the factor and the simulated time are in every frame's header); the header names the assistance, the caption reads the truck's or the control's signals. Every film was looked at in contact sheets of four frames and the figures' frames in full; what they show, faults included:

ASSISTED ROBOT FIXTURES (E14b; a close-up of the working hand or foot where one exists and a cab camera):

- `robot_lift`, `robot_tilt`, `robot_shift`: the right index finger on each cowl lever's handle, pulled and then pushed through its travel while the caption reads the lever angle, the valve signal and the mast's response; the other levers stay still.
- `robot_accel`, `robot_brake`, `robot_inch`: both cab cameras (the pads are under the dash from every position tried): the shin lifting and pressing, the caption reading the pedal's travel (the brake to 0.96, held at half, released) and the foot's contact force; in `robot_inch` the pedal never moves.
- `robot_direction`, `robot_horn`, `robot_park`, `robot_key`, `robot_steering`: the left index finger working the stalk through F, N, R, N; the fingertips on the horn button; the parking-brake release (the close-up is mostly the robot's own trunk and arm, the cab camera shows the lean and the reach); the key attempt (the close-up is blocked by the reaching right arm; the key never turns); the left hand on the spinner knob carried round and regripped (the close-up from behind the wheel is dark; the caption shows the wheel angle falling back on each regrip).

ASSISTED task demonstrations (a third-person camera that picks a clear placement each moment, and the robot's head camera, the policy's view):

- Warehouse: `task_start_and_park` (to the parking bay at the wall, the forks lowered and tilted to the floor), `task_aisle_cruise`, `task_drive_to_pallet`, `task_carry_to_floor_bay` (the wrapped load set down in bay B-03), `task_carry_loose_stack`, `task_put_away_rack` (the load lifted to the second beam level and set on the beams; Figure 15), `task_feed_conveyor` (the coil pallet set on the roller conveyor between its guides, then carried away toward the press). In `task_drive_to_pallet` a column passes between the camera and the truck for about a second.

- Dock: `task_load_trailer` (over the leveler, the pallet set behind the nose pallets, out again; Figure 16), `task_load_trailer_chocked`, `task_load_bales` (two bales set side by side on the dunnage strips at the nose), `task_unload_trailer`. Inside the trailers the head camera's view is dim but readable (walls, floor, the dash), metered as the D435i would.
- Yard: `task_yard_route` (the marked lane to the foot of the ramp), `task_ramp_up_loaded` (up the yellow ramp into the trailer; the head camera inside is well exposed, about 90 of 255), `task_ramp_down_loaded` (reversing out of the trailer, down the ramp and to the drop zone; leaving the trailer the head camera is washed out for a moment while its exposure adapts to daylight).

Predeclared negative controls (section 11): `negative_no_parking_brake`, `negative_forks_high` (squared up to the pallet with the blades 85 mm up, read in the caption), `negative_fast_turn`, `negative_stack_hard_stop` (from beside and behind: the cartons stay on the pallet), `negative_chocked_hard_stop`, `negative_ramp_too_fast` (coming out of the trailer the head camera is washed out for a moment while its exposure adapts to daylight).

Physics experiments: the truck with the 67 kg seat ballast standing in for the operator unless stated: `physics_lift_rated_load` (E5: the rated load raised to full height and lowered, the caption reading the lever, the stage pressures and the steer axle's load), `physics_full_brake_laden` (E8), `physics_pallet_full_brake` (E9: the 800 kg load on the pallet holding on the forks; the forks camera shows the pallet's deck at the blades), `physics_stack_full_brake` (E9: the loose carton stack under a full brake, the top cartons staying on), `physics_hill_start` (E12: up the 13.3 % ramp slab from a standstill), `physics_runup_turn` (E7: the run-up and the turn at 30° of lock, the caption reading the speed, the peak roll and the four tyre loads, the inner front one falling to zero), `physics_seated_ride` and `physics_seated_stop` (E13, with the H1-2 seated: its trunk pitching toward the wheel in the stops, the caption reading the hip point and the pelvis frame on the cushion), `physics_column_strike` (E10: the truck driven into a column guard at 4 km/h, the fork tips meeting the guard and the truck stopped by it) and `physics_legal_abuse` (E17 seed 0, the H1-2 seated: the robot thrashing under random legal commands, its limbs flung across and out of the cab, the caption reading its fastest link, the seat switch and the solver's gradient).

Start states at rest (E0, 120 s at 10×): `rest_put_away_rack`, `rest_load_trailer_chocked`, `rest_ramp_up_loaded`: nothing moves.

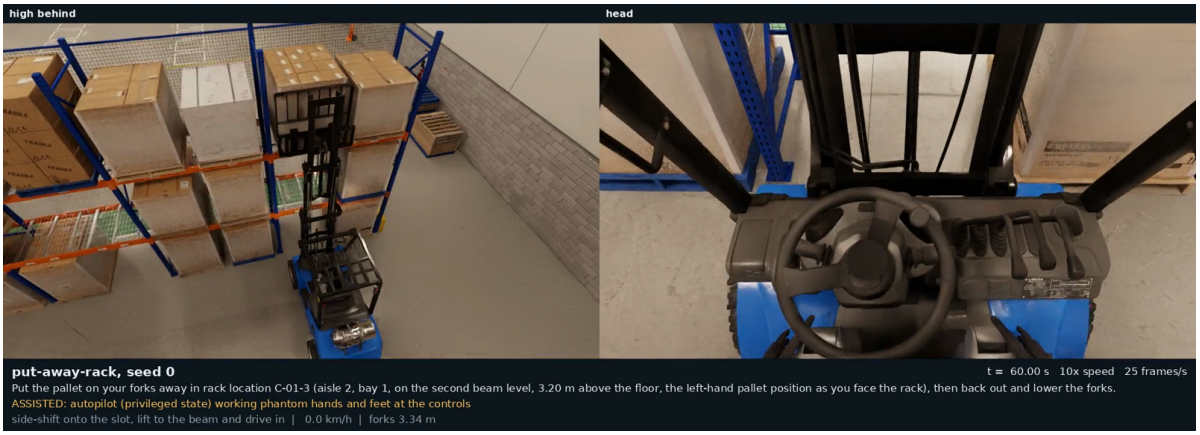


Figure 15. From the film task_put_away_rack (an earlier build): the load raised to the second beam level (forks at 3.34 m) and driven into its slot, seen from high behind, and the policy's head camera, with the task, the assistance label, the phase, the speed and the fork height.

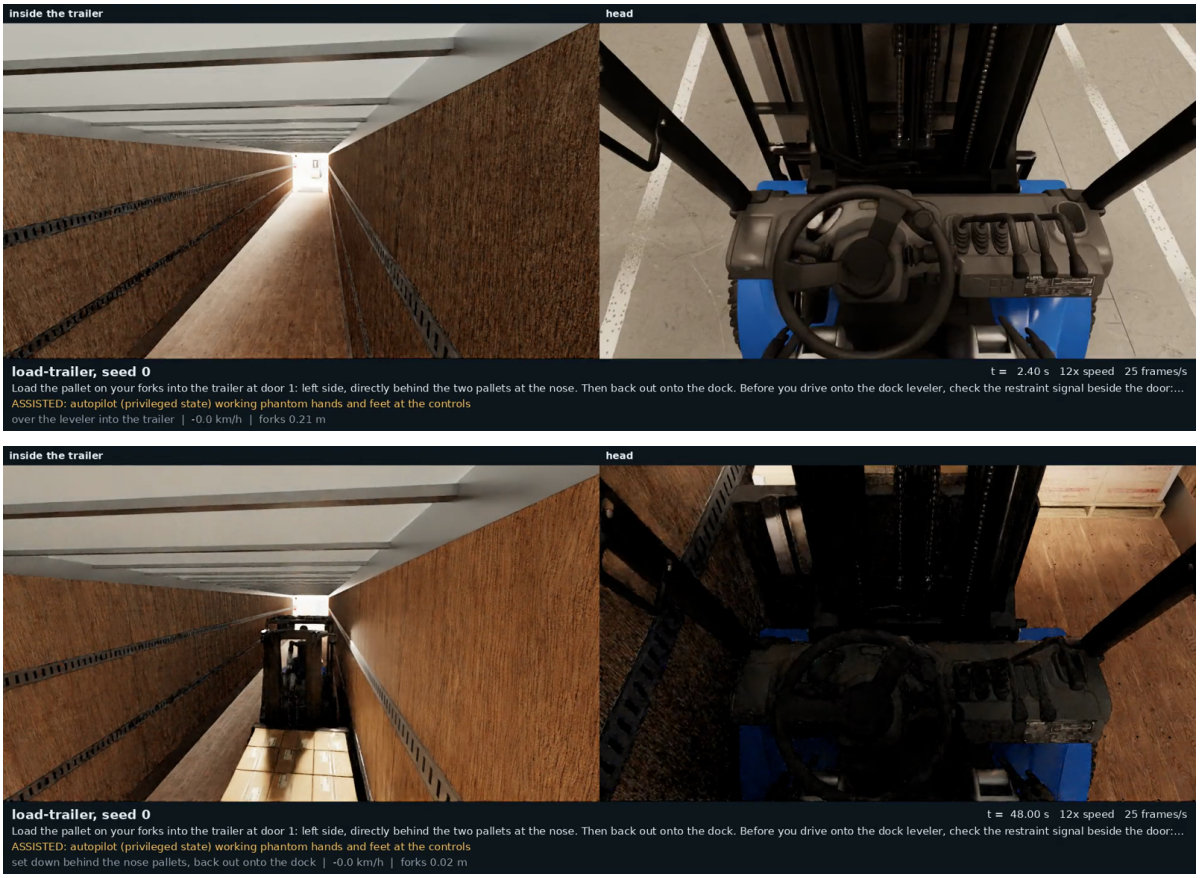


Figure 16. From the film task_load_trailer (an earlier build): on the dock approaching the leveler (top) and deep inside the trailer, the pallet set down behind the nose pallets (bottom), where the head camera, metering its own frames, still shows the walls, the floor and the dash.

15. Limitations and work not built

What the environment does not do, and where the evidence stops short, measured where it can be. None of these is absorbed by the evaluator; several make tasks harder, none makes them easier.

The parking brake and the seated humanoid (the most consequential open item). Every task starts with the parking brake applied (`task.py`). In an assisted fixture of an earlier build (E14b, section 10) the seated H1-2 released it with its left index finger's pad under the handle's end (135 N, 14.9 N·m about the hinge), its pelvis held by the external support; it did not apply it again, because the handle's apply side lies inside the asset's cowl when the lever is released, so a hand reaches only its underside, which pushes toward released. Without assistance the scripted H1-2 has not released it: inverse kinematics reaches the handle once the torso joins the reach, but the H1-2 has no waist pitch, and leaning in brings its torso onto the spinner knob and its right shoulder onto the tilt lever, so the palm stops about 58 mm short (section 10). A policy may find another way (a forearm, the back of the hand, a sideways reach); none is demonstrated. Whether the asset's lever and cowl stand where the real H2.5A's do was not verified (no cab drawing was found); the lever was not moved, because moving it to suit the robot would bend the world. Cause: open between the robot (its trunk cannot flex), the fixture planning and the asset. The driving-robot kit releases and sets it by contact (E14k).

Braking with the rated load (E8, section 9.5). The brake factor is the smallest that meets the standard, and still a laden stop at the standard's 450 N lifts the steer wheels for 0.57 s, past the evaluator's 0.3 s wheel-lift rule; a policy carrying the rated load at the rated moment must brake more gently than that. The real truck's brake factor and its margin over the standard are not published (open); with 20 % more brake the same stop lifts the wheels for 1.89 s.

Lever slams with the rated load (E5, section 9.3). A smooth rated-load cycle to full height keeps the steer

axle within the 0.3 s rule (0.14 s at most), but slamming the lift lever open and shut (within 0.2 s) with the rated load lifts it for up to 0.36 s; the abrupt cycles pitch the truck at most 1.6° and none overturns it. That is the physics of stopping 2.8 t moving at 0.6 m/s, which is why operators feather the levers; the evaluator's wheel-lift rule then fails the episode. Whether the H2.5A's lowering regulator is load-compensated (as modelled) or a restrictor type is not published; with a restrictor the smooth cycle's lowering would empty the steer axle for 0.22-0.35 s (E5b). The tyres' hysteresis damping sets the pitch mode's damping (ζ 0.024-0.031 at 0.83 Hz, against one measured forklift's 0.031); it is fitted to an 18x7-8 tyre's measurement, not measured on these tyres.

Violent legal commands at the low end of the fingers' inertia (E17). At the ledger's finger armature the world survives eight 60 s seeds of violent legal commands with no fault (section 12.3). The value is the middle of an open range (0.003-0.03 kg·m^2 per finger); in an earlier build, at the range's low end one of two seeds faulted at 6.1 s with a thumb joint at 105 rad/s, and that run was not repeated. The finger drive is modelled as a joint back-driven through its rotor inertia; a lead-screw drive is expected to resist back-driving (an expectation, not a measurement), so a model of the drive's back-driving is the cause-level next step. Cause: open (the finger drive's model).

The robot as an operator (SCRIPTED ROBOT, section 10). The scripted operator is a demonstration tool, not a policy; where it stops, the cause is stated and whether the world or the script owns it.

- Without assistance the script passes no control by its own rule: the lift lever reaches 88 % and 85 % of the valve's travel (95 % required), the brake pedal 0.47 before the foot leaves the pad, the accelerator 0.45, and the inching pedal does not move under the foot; multi-turn steering by knob regrip or rim fails, the direction stalk and parking brake do not move. Its foot and hand placement (28-86 mm off on the pedals) and its lack of trunk posture control (the trunk folds forward as a foot lifts) are the first limits. Cause: the script.
- With assistance (E14b) the lift and tilt levers, the accelerator and the service brake pass every criterion on the current build, and the direction stalk and horn did in an earlier build. The side-shift lever is pulled 1 % short of its rule; the key (the fingertip never reached it), the inching pedal (0.50 at most), the steering's return from the left lock and the parking brake's re-apply (above) do not pass. Cause: the fixture's planning for the key, the inching pedal and the steering.

The driving-robot kit (section 4.4). At the products' published forces the brake robot takes the pedal to 78 % and the inching robot to 87 % of their travel: the brake's hydraulic floor stop needs 1,800 N and the inching pedal carries the brake's force past 17 % of its travel. The brake press already gives 1.5 times the standard's line pressure. The lever-bank figures are a gearshift robot's and no hand-brake actuator product was found; the key and horn robots' limits are estimates. The kit has driven three task runs (two tasks); the other twelve tasks, a cold render-worker start and any film of the kit are not measured.

The seated ride (E13, section 12.2). The passive H1-2 stays on the seat in hard stops with the brake held at 1.2 times the standard's torque (hip point within 20 mm), but its trunk pitches forward 11.5° on its hips and the pelvis frame moves 39 mm, because the robot holds its start command and nothing braces it; a seated human braces with the feet, the hands and the back, and a policy must too. The belt's retractor locks in these stops and the head stays 27 mm under the guard. The cushion is a rigid block at its loaded thickness with μ 0.6 (the robot's shell, at μ 1.0, sets the grip under MuJoCo's rule); a foam bed was tried and changed little; a deformable seat that forms a pocket around the body is not built.

Contact fidelity at the forks. In this world MuJoCo placed a convex piece's contacts on its centre line (a standalone MuJoCo model with the same shapes gave contacts at the ends; the difference was not found), so the fork blades are eight box pieces and a load's support ends 62 mm short of the tip taper. A load fully on the forks tips forward at about 2 m/s^2 where the real blade would hold it to about 3 m/s^2 (computed from the support polygon). Loads are convex meshes because MuJoCo's box-box routine produced garbage contacts four times, the last between two small cartons (section 5).

The cab's collision geometry. The cab interior is a CoACD decomposition (concavity threshold 0.03) carved round the key, the parking-brake lever's slot and two open spaces under the left housing (138 hulls, each carved hull inside an uncarved one of the 60), plus authored footwell plates. At the robot's measured contacts the hulls stand 6-11 mm proud of the asset's surface elsewhere; a finer decomposition (threshold 0.015, 160 hulls) lowered the median deviation only to 13.6 mm and was not adopted.

The truck model.

- Drivetrain: laden 0-15 m in 5.15 s against the sheet's 5.9 s (a held-out prediction missed by 13 %; the converter and engine curves were fitted to the drawbar pull and the unladen top speed only, and were not refitted to the held-out number). Cause: the world (the converter-engine model).
- Gradeability: the sheet's 4.8 km/h on 28 % laden and 26 % unladen is missed both ways: 0.84 km/h laden with the engine lugged to 1,784 rpm, 6.69 km/h unladen. The sheet's figures are one number (its 28 % is the drawbar pull over the laden weight), so the held-out check is weak, and E21b shows the tyres' friction does not move either result much. Cause: open (the converter's curve at speed).
- Tyre friction: dry concrete is taken as SUPREM's drum (no measurement of a super-elastic tyre on a warehouse floor was found), the longitudinal law rests on one tyre's braking-slip curve at one load, and the other surfaces are estimated ratios. E21b brackets the level and the load law.
- Hydraulics: raising the rated load, the closed lift column's pressure reaches 19.0 MPa against the pump's 18.5 MPa relief (the column's pressure is its load's, and the relief sits at the pump), and 19.8 MPa while the rated load is lowered; the relief is a force cap at the pump, not a valve with its own dynamics.
- The centre-of-gravity height is derived from the sheet's axle loads and the asset's geometry, not measured; the stability results (E6, E7) inherit it.
- Trailer walk: the chocked trailer settles 3.3-3.5 mm from its spot after each hard stop near its nose and does not accumulate over five entries (section 9.8): dock walk, which real air-ride trailers show under repeated laden entries (Hendrickson L816), is not produced by this model. The trailing-arm suspension's fore-aft travel is modelled with an open slope; what makes a real trailer ratchet (slip of the tyres or sand shoes between entries, the suspension's hysteresis) is not. With the spring brakes set, the chocks carry no force in a single stop (E11), so chocks and no chocks give the same movement; whether real chocks engage under a braked trailer was not established. Cause: open (the trailer's support model).
- The E11 probes' own driver let the carried pallet's deck strike the trailer floor in every stop (26-65 kN at 0.24-0.47 m/s); their verdicts on the trailer's movement stand, but the probe's driving is not a careful operator's.
- The lowered forks rest on the floor in a parked start state (the tyres then carry 274 kg less: carriage, side-shifter and forks), as on a real truck; E0's axle loads are therefore not the sheet's.

Tyres. The super-elastic tyre model was validated inside the truck (E3, E6-E8, E12's coast-down within 0.11 %) but not on a single-wheel rig against the SUPREM curves, and no comparison against MuJoCo's native tyre contacts was run.

The asset's width. The Forklift C asset is 1.322 m across its front tyres and 1.364 m across its rear (the H2.5A is 1.16 m); the environment keeps the asset's geometry, and the trailer side-wall threshold is derived from it (125 mm). The asset's outside turning radius, 2.365 m, is about 4.5 % over the sheet's, and the rack aisles are 0.17 m wider than the VDI formula for it.

Rendering. Films are rendered after the run from its recording (every body's pose at 25 Hz) by the render-only worker at 25 frames per second, each film's speed chosen so that it lasts about ten seconds (2× to 10× real time; the task's sim seconds are in each frame's header); every camera meters its own frames (the D435i's exposure range); the policy camera is rendered live at each observation. The films and stills of section 14 come from an earlier build's recordings, and no film of the kit exists. The dash overhangs the pedal pads from every camera position tried, so no film shows a sole on a pad; the pedals camera shows the leg and the caption the pedal's travel. Inside a trailer the head camera's image is dim but readable: deep in the dock trailer the film's head tile averages 27-45 of 255 (median 8-22) and in the yard trailer about 90; the carried load hides what lies ahead, as it does for a seated driver. Rendering under GPU pressure (four render workers sharing the 32 GB GPU with other sessions) can produce empty frames or crash the worker; such a film is rendered again.

Runtime. Through the served interface the H1-2's commands of 20 ms or longer and the kit's of 0.1 s or longer stayed inside the 200× limit at their 99th percentile, with no request over it (section 13); shorter commands do not always, because each request's fixed cost and new camera frames dominate so short a command, and stalls of the shared host are not bounded by anything in the environment. A request after a pause longer than the render worker's 600 s release waits 71-99 s for a new worker. All on a shared host whose load and GPU were busy throughout; no quiet window opened, so no quiet-host figure exists.

Scope. Not built: pedestrians and other traffic, the pre-use inspection, refuelling (the LPG cylinder change), the seat belt's buckle (the belt is fastened at the start and cannot be unfastened), and any policy.

16. Reproduction

Software: Newton 1.5.0, Warp 1.16.0, MuJoCo 3.11.0 and mujoco_warp 3.11.0 (physics, CPU); Isaac Sim 5.1 (render-only worker, GPU); Blender 4.5.3 for CPU debug renders; CoACD 1.0.5 (pinned in `assets/install.py`) for the cab decomposition. Assets: `python -m forklift.assets.install` downloads the NVIDIA Forklift C, the Simple_Warehouse props and the sky HDR and checks each against `assets/assets.lock.json` (size and SHA-256). The H1-2 with Inspire hands comes from the shared robot catalogue (unitree_ros at 5994d4faef0a9cadd3287f8de0199a67eeb2a259, BSD-3-Clause). The driving-robot kit is an authored catalogue model with its own Newton builder and needs no download.

The environment runs through the GAR CLI from the release:

```
python -m forklift.assets.install                      # 2,290 hash-locked files
python -m robots.models.h1_2.install                   # the H1-2 with Inspire hands
gar doctor forklift                                    # assets, hashes, robot builders, renderer
gar tasks inspect forklift                             # the fourteen tasks and their selectors
gar run forklift --select task=put-away-rack --select seed=0 --policy-key
gar run forklift --robot drive-kit --select task=start-and-park --select seed=0 --policy-key
python -m forklift.manual --session SESSION_ID         # manual control page (policy key)
python -m forklift.geometry --check                    # the shipped cab decomposition against its lock
python -m forklift.variants --freeze                   # data/variations.json from the resolver
```

A failed `gar doctor` row names the command that fixes it. The scripts that ran the experiments, fixtures and demonstrations of sections 9 to 14 and rendered the films are not part of the release.

17. Sources

Labels: read (opened and read), read* (through a fetch summariser), search (a search result's summary), measured (from the asset or a probe here), derived, estimate, open.

- Truck: Hyster H2.0-3.5A technical guide rev03 (H2.5A LPG; masses, axle loads, speeds, gradeability, drawbar pull, turning radius) [read]; Toyota 8FG/8FD25 specifications [read]; Danfoss steering units (OSPC, LS units, general steering components) [read]; Bolzoni Auramo fork guide [read]; Lift-Tek three-stage mast manual [read]; Hangcha XC maintenance manual [read*]. NVIDIA Isaac Sim 5.1 Forklift C asset [measured].
- Tyres: SUPREM, "Empirical Lateral-Force-Model for Forklift Tires", Logistics Research 10:1 (2017), DOI 10.23773/2017_1 [read]; Pross and Bruns, Logistics Journal: Proceedings 2019 (braking-slip and lateral curves of super-elastic tyres on the same drum) [read, digitised]; Oh et al. 2013 (load-deflection and damping of an 18x7-8 super-elastic tyre) [read]; "Virtual Testing of Counterbalance Forklift Trucks", Machines 8 (2020) 26 (Traigo vertical law) [read]; Trelleborg Elite XP catalogue [read]; Wong, *Theory of Ground Vehicles*, via the HPWizard tyre-friction page [read]; Haritos, Nguyen, Gad and Wilson, "Floor vibrations induced by forklifts", AEES 2013 (a forklift's measured pitch damping) [read].
- Brakes, inching, steering and hydraulics: TCM FG20T3/FG25T3 (2.0-3.5 t) service manual SEF-0F7BE (brake hardware, pedal heights and plays, shoe clearance) [read]; OSHA standard interpretation of 1990-06-21 (B56.1's 25 % brake drawbar drag) [read]; GB/T 18849-2023, the identical adoption of ISO 6292:2020 (450 N pedal operating force, 1,800 N fully pressed) [read*]; Komatsu US4441596A (inching pedal working the brake master cylinder) [read*]; Eaton US 6,769,249 B2 (steering slip of 2-5 rpm at the travel limit) [read]; Toyota Industries EP2962981A1 (pressure-compensated lowering flow control) [read*].
- Rules and stability: OSHA 29 CFR 1910.178 and Appendix A, the OSHA powered-industrial-truck eTool (load handling, ramps and grades) [read]; NIOSH Alert 2001-109 [read]; INRS TI-NT-46 (EN 16203 tests) [read]; ISO 3691-1:2011 sample [read]; ITSDF B56.1 interpretations [read].

- Facilities and loads: GMA stringer pallet (NWPCA sheet) [read]; Steel King rack design guide [read]; Hyster H2.0-3.5A technical guide's aisle drawing (the VDI 2198 right-angle stacking aisle formula), Cisco-Eagle and Toyota aisle-width guides [read]; Interlake Mecalux selective pallet rack guide and Cisco-Eagle's rack clearance guide (the 6 in lift-off clearance) [read, read*]; Blue Giant dock height and leveler sizing [read]; Hendrickson L816 (trailer sag, walk) [read]; Wabash DuraPlate HD inside dimensions [read]; Bluff MobilePRO yard ramp [read]; US10988360B2 (steel-on-wood friction) [read]; OCC bale dimensions [search].
- Lighting: IES recommended illuminances for warehouses (10 fc active bulky storage, 30 fc small items and shipping and receiving, 30-50 fc at docks) [search: the Lighting Design Lab footcandle guide and lighting-vendor summaries]; a 15,000 lm 150 W high bay (Lithonia IBG 15L, 2 x 1 ft) [search].
- Operator, seat and guard: ISO/TS 15066 Table A.2 (130 N for the skull and forehead), through its reproduction in arXiv:2311.13814 [read]; OAR 437-002-0227 (39 in seat-to-guard minimum) [search]; Hyster J2.2-3.5XN technical guide (80 mm seat suspension travel) [read]; NIOSH (Mayton et al., seat suspension measurements) [read]; UN R16 (belt retractor locking: vehicle 0.45 g, webbing, retraction) and FMVSS 209 [read]; Grammer MSG 20 sheet [read]; ISO 11226 / EN 1005-4 trunk posture band, through Continental AG's ergonomics requirement specification (2024) [read].
- Robot: Unitree H1-2 and the Inspire RH56 hand (URDF and meshes from unitree_ros) [measured]; the H1-2 product page [read].
- Driving-robot kit: AB Dynamics steering robots specification (SR 15 Orbit), pedal robots specification (Accelerator Robot, RBR 600) and CBAR 600 page [read]; 4activeDR brochure (brake actuator, ADAS/NCAP configuration) [read]; GSA9001S automatic gearshift control (exhibitor brochure) [search]; Staehle actuator catalogue (ignition key and start/stop button actuators, no figures) [read].